

claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf\_f9aa1d1c-c83\agent-ae40e675b74afcf13.jsonl`
- SHA-256: `dd4d1e8f75756d85e6a6fa0de4e1546c1e7768d9c144568a9279699e97f6e57d`
- Source modified: `2026-06-09T11:59:04+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf\_f9aa1d1c-c83`
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`

Transcript

1. user (2026-06-09T11:56:27.054Z)

Completeness critic. Given the recommendations + research, list: gaps, any guardrail (MIT/Apache/BSD; no-paid-LLM-training) misalignment, weakly-sourced claims, and anything that contradicts the owner's reframed vision (quality-first / on-device-later / multisport-single-conversational-model). Concise.

RECS:

{"q1_model_and_starting_point":"CATEGORY FIX FIRST (the Gemma-4-vs-ms-swift confusion): these are NOT alternatives. ms-swift is a FRAMEWORK (the \"forge\" ? the trainer that runs SFT/GRPO on a GPU and emits LoRA/full weights). Gemma 4 / Qwen3-VL are BASE MODELS (the metal you forge). You pick ONE framework and feed it ANY Apache/MIT base; swapping the base later (Qwen->Gemma) is a config change, not a re-platforming. So the decision is two orthogonal picks: (a) framework, (b) base ? plus (c) what KIND of model to start with.\n\nRECOMMENDED FRAMEWORK: ms-swift (Apache-2.0). It is the only one of the five leading forges (ms-swift / LLaMA-Factory+EasyR1 / unsloth / TRL / axolotl) that clears all four hard requirements in ONE stack: native video training, native audio, timestamp/grounding controls (fps/num_frames + JSON-messages timestamps), AND GRPO/RFT with a CUSTOM external-reward plugin over VIDEO rollouts. Verified gaps in competitors: LLaMA-Factory offloads GRPO to EasyR1 which is image-only; unsloth's Vision-RL and TRL's GRPO-VLM are image-only; axolotl's video path is self-described \"not well tested.\" All five are permissively licensed, so the commercial guardrail is NOT decided at the framework layer ? it is decided by base WEIGHTS + DATA. Keep unsloth in the back pocket as the 8GB-local SFT/compression tool for the DEFERRED on-device goal (note: unsloth SFT *does* now do Qwen3-VL video; only its RL path is image-only ? don't over-read 'image-only'). This hardens the lean already in docs/PLATFORM_ARCHITECTURE.md (which favored ms-swift but predated the RFT-reward + audio requirements).\n\nRECOMMENDED BASE FAMILY (when you get to the VLM): Qwen3-VL (Apache-2.0, uniform across 2B/4B/8B/32B/30B-A3B/235B ? verified no per-size fork, unlike Qwen2.5 which forked at 3B/72B). It is the only candidate hitting all four base-model requirements: clean Apache-2.0 weights, native long-video (256K->1M context = full match, not a 60s window), purpose-built timestamp emission ('Text-Timestamp Alignment' for second-level localization), and multi-turn video chat. Update your docs:\n\nOWNED_MODEL_TRAINING_STUDY.md/IMPLEMENTATION_PLAN.md still name Qwen2.5-VL-7B primary and call Gemma-4 'AMBER' ? verification confirms Gemma 4 shipped April 2026 under genuine standard Apache-2.0 (clear the AMBER), BUT it caps video at ~60s with NO timestamp output, so it is DISQUALIFIED as the primary extractor; keep it on the bench ONLY for its native audio (racquet-hit cues) if a fusion signal is later wanted. InternVL3 (MIT+Apache, no native timestamps) is the license-clean fallback if a Qwen blocker appears. Reject Gemma 3 (viral), Gemma 3n (Gemma Terms FOLLOW DERIVATIVES ? even cleaner reject), Llama 4 (700M-MAU cap). (Minor: Qwen3-VL weights shipped Sept-Oct 2025, not 'Nov 2025' ? that's the arXiv report date; fix if it appears in a shipped doc.)\n\nSTARTING POINT ? START WITH THE TAL HEAD, NOT THE VLM, NOT BOTH-AT-ONCE. The evidence is decisive that a fine-tuned VLM does NOT reach 'very high' precision/recall at strict temporal boundaries: the strongest RL-post-trained video VLM (Time-R1 on Qwen2.5-VL-7B) hits only R@0.7=50.1% / mIoU=60.9% on Charades-STA even fine-tuned; the literature is explicit that VLMs 'capture coarse temporal regions rather than accurate event boundaries.' Specialized TAL heads still lead at the boundary (ActionFormer ~66.8% mAP THUMOS14; TBT-Former 68.0%, Dec-2025). Boundary accuracy IS the product. And the head is ALREADY de-risked on YOUR corpus

(rally_seq_proto.py: per-frame AUC 0.84-0.91 held-out, LOVO 0.583-0.615), trains in minutes at ~4.5GB VRAM, \$0 on the 2070 Super, is ~1M params (trivially compressible later), and DOUBLES as a confidence-gated pseudo-label teacher AND the honest baseline the VLM must beat. So: this is BOTH, strictly SEQUENCED ? head now (quality-first extractor), VLM later (deferred conversational/grounding upgrade), which is a sharpened version of your existing Gen-0->Gen-2 blueprint. Concrete first 2 experiments: (1) M0.4 in OWNED_MODEL_IMPLEMENTATION_PLAN.md ? productionize rally_seq_proto.py into a one-command train->eval->ship-gate harness over ActionFormer (MIT, 1:1 rally=instance) and/or ASFormer (MIT, TAS), reusing metrics.match_intervals as the single scorer; ship-gate = held-out segment F1 beats the honest heuristic LOVO (~0.54). (2) M0.2 ? grow n=3 -> n>=5 distinct matches (the bottleneck is data+calibration, not method); score each as it lands. Both run on the local GPU/WSL box, need no cloud, no DPA. The VLM (Gen-2) stays gated behind platform scaffolding + healthy corpus + explicit go; when it comes, prefer the two-stage pattern (VLM coarse-proposes, head refines boundaries ? TimeRefine) over wholesale replacement, and originate your own grounding-CoT supervision because the public CoT datasets (TVG-R1's 56K) are Gemini-generated and thus FORBIDDEN here.", "q2_conversational_qa": "RECOMMENDATION: PUNT the conversational-QA model/feature; BUILD the bookkeeping now. These are separable and both halves are decisive.\n\nPUNT THE FEATURE (revisit after rally/highlight quality is solid). Quality gates everything: your honest cross-video LOVO is ~0.583, far from 'very high P/R.' Conversational answers are only as good as the extraction underneath, so shipping QA now would expose a weak extractor through a chat UI. There is no quality cost to waiting and a large one to rushing. Also reject the end-to-end-VLM-answers-each-turn architecture outright: it is expensive, bad at exact counting ('rallies over 20 shots'), and the strongest teachers you'd want to distill from are license-forbidden.\n\nBUILD THE BOOKKEEPING NOW (no-regret, and the brief requires it). The architecture is 'extract once, query many': EXTRACTION (the TAL head, later optionally a VLM) emits structured timestamped spans -> a durable PER-VIDEO EVENT STORE -> a cheap QUERY LAYER answers. Critically, the queries the owner cited ('longest rally', 'all service faults', 'rallies over 20 shots') are STRUCTURED-NUMERIC, not semantic ? they are text-to-SQL + a clip-cut (ffmpeg/VideoCompiler), NOT a job for a VLM. SQL is exact where VLMs hallucinate counts. And MOST OF THE STORE ALREADY EXISTS in this repo: I verified backend/infrastructure/database.py has play_segments + events + candidate_segments tables on a PRAGMA user_version migration ladder, and metrics.match_intervals is the scorer spine. So the bookkeeping is an extension, not a greenfield build.\n\nARCHITECTURE (extraction -> event store -> query layer), designed now even though QA ships later:\n1) EXTRACTION writes the per-video event index. Make the event-index SCHEMA a first-class data contract alongside the M0.1 labeled-corpus spine: per rally {video_id (file MD5), rally_ordinal, sport, start, end, shot_count, ending_reason/fault_tags, derived_clip_ref, provenance, confidence}. This is the same row the M0.1 transcoders already produce ? wire it to a queryable store, don't invent a parallel one.\n2) EVENT STORE = the existing SQLite tables, extended (highlights becomes first-class per PLATFORM \$4.1; add the rally-inventory/event-tag columns). Postgres later via the planned repository layer; no architectural change.\n3) QUERY LAYER = a structured-query CONTRACT (a small set of parameterized SQL queries + clip-cut) NOW; the natural-language parser/agent is the part you PUNT until quality clears the bar. Paid Gemini stays inference/fallback ONLY (never a trainer) for genuinely open-ended questions at the edge ? allowed because it reasons at inference over YOUR structured index, it is not trained on its outputs.\n\nOne caveat to flag: richer semantics ('service fault') require explicit event types the head/labels must emit, so the label taxonomy in M0.1 must reserve those event/fault tags up front even if you don't populate them yet. The single concrete action now: fix the event-index/query CONTRACT early (schema + the parameterized query set) so extraction and the future QA layer are built against a stable seam.", "q3_multisport_impact": "Multisport-from-day-one is FEASIBLE as a single sport-agnostic model and even beneficial ? but it does NOT change the model/framework recommendations; it changes the DATA and DETECTOR work, and it adds one real licensing front. Break it down:\n\nSINGLE SPORT-AGNOSTIC MODEL IS FEASIBLE: yes, for both the TAL head and the eventual VLM. The head is sport-agnostic by construction ? sport is just another input feature or a per-sport calibration, one shared head, not per-sport models. The VLM base (Qwen3-VL) is a general model fine-tuned across all racquet sports; one base, many sports, no per-sport bases. The collector + indexer are already sport-generic by design and the repo rename off 'badminton' is consistent with this. CAVEAT ON THE EVIDENCE I MUST FLAG (verification correction): the synthesis cited RacketVision (AAAI'26) as proving unified multi-sport TEMPORAL training boosts generalization +15-19% mAP. Adversarial verification REFUTED that framing ? RacketVision is a ball-TRACKING benchmark, not a temporal-localization one. So treat 'unified multi-sport training improves temporal generalization' as an OPEN HYPOTHESIS, not established fact. The single-model concept remains sound on first principles (the rally signal ? trajectory visibility/velocity/net-crossing ? is a racquet-sport invariant), but it is asserted-not-proven for temporal localization. Don't cite RacketVision as evidence in shipped

docs.\n\nMODEL/Framework: UNCHANGED. ms-swift trains any sport's data; Qwen3-VL is sport-agnostic; the TAL head is shared. Nothing in multisport breaks these picks.\n\nDATA / COMPUTE / STORAGE: scale UP, as the owner accepts. More labels (each new sport needs its own labeled matches ? split by match), more compute (more training data, more pseudo-labeling bursts), more storage (more corpora). The accepted 'good problem.' Real bound is not the model ? it is solo-founder labeling throughput + per-job cloud spend, so sequence sports (badminton -> tennis/TT, where public datasets/detectors are richest -> pickleball/padel, which have far thinner public coverage) rather than boiling the ocean. The current de-risking corpus is badminton-only; the feature stack and the only labeled matches are badminton, so multisport generalization of THIS stack is genuinely unproven and is the honest gap.\n\nTHE NEW LICENSING/ENGINEERING FRONT (the binding multisport blocker): per-sport BALL/SHUTTLE TRAJECTORY DETECTORS. The head's features come from a trajectory detector. Today that's WASB (MIT code) for badminton ? but its distributed checkpoint has UNRESOLVED provenance (the C3 blocker), a commercial-cleanliness issue that must be resolved (own-train weights or swap) before commercial ship; a clean model trained on provenance-unclear features does not launder the provenance. Multisport REOPENS this per sport: clean MIT/Apache/BSD trajectory detectors for tennis/TT/pickleball/padel are NOT yet identified. So the multisport blocker is detectors + per-sport labeled data + the WASB C3 resolution ? NOT the model architecture. Net: keep one shared head and one base; sequence the per-sport data + detector cleanliness work; carry WASB C3 as a Gen-0 prerequisite.", "q4_language": "Stay PYTHON everywhere for the foreseeable roadmap; treat 'another language' as a DEFERRED, layer-local DEPLOYMENT detail, never a rewrite and never a hot-path split. Each of the three layers independently points to Python or to a choice structurally decoupled from your dev language:\n\n(1) ML TRAINING + EVAL: effectively Python-locked, and that is fine. The PyTorch/Transformers/ms-swift/ActionFormer/ASFormer stack is Python; ms-swift is invoked as a CLI subprocess (swift sft/eval), so even a non-Python orchestrator would just shell out identically. No non-Python option retains native-video VLM training or your dependency-free numpy LogisticRegression Gen-0 seed. Locking costs nothing because training is offline/batch, not a latency hot path.\n\n(2) PLATFORM (async jobs, API, storage/compute backends): no real Python 'hot path' to escape. The heavy bytes/GPU work already runs OUT-OF-PROCESS (ffmpeg Windows-side, WASB in WSL, cloud GPU per-job via Modal/RunPod) ? Python is the conductor, not the orchestra. The control plane is I/O-bound (DB, signed URLs, job status, polling) where the GIL never bound you, and Python 3.14 free-threading + FastAPI 0.136 free-threaded support remove the residual CPU objection. The Python-native pieces already chosen in PLATFORM_ARCHITECTURE.md (fsspec/smart_open, Arq->Hatchet, Postgres + repository layer) cover it. A Go/Rust control-plane service would buy single-digit-ms wins on requests dwarfed by 40-minute GPU jobs ? pure overhead for a solo founder, and it would FRACTURE the one codebase that must hold per-video QA state + eval harness + provider registry together. REJECT the Go/Rust hot-path split.\n\n(3) ON-DEVICE (the DEFERRED upload-cost-saver): the ONLY layer where a second language eventually appears ? and it does NOT dictate your dev language. You train in Python and export across the EXPORT BOUNDARY to ONNX/Core ML/ExecuTorch/GGUF; the device host (Swift+Core ML for iOS, C++/Kotlin+ExecuTorch for Android, Rust via the ort crate for desktop) runs the artifact with zero Python dependency. So on-device dictates a thin, separate, much-later runtime SHELL ? written only if/when a phone model ships. Per the owner's reframe, on-device is explicitly LATER, so this is the correct layer to punt, and punting costs nothing because the export boundary IS the seam.\n\nThe conversational-QA + per-video bookkeeping reframe REINFORCES Python-everywhere: durable relational store (SQLite->Postgres) + a thin query/agent layer is bread-and-butter FastAPI/Python; splitting languages would fracture the state layer.\n\nTHE ONE CONCRETE LANGUAGE ACTION NOW: none on the dev language ? but impose one constraint on the MODEL: train with export-clean (ONNX/Core ML-friendly) ops so the eventual compression target stays reachable without re-architecting. Escalation order if a real Python CPU hot path ever appears under load: free-threading -> numpy/vectorize -> a Cython/PyO3 C-extension (still a Python module) -> only then a separate Go/Rust service. No such hot path exists today. (Licensing note: the convenient on-device default 'Gemma 3 270M GGUF' is REJECTED by your guardrail ? keep on-device bases on the Qwen3-VL-4B Apache-2.0 lineage.)", "roadmap_implications": ["FRAMEWORK DECISION: ratify ms-swift (Apache-2.0) as the single SFT+RFT forge; keep unsloth as the secondary 8GB-local SFT/compression tool for the deferred on-device goal. This hardens (does not contradict) the lean already in PLATFORM_ARCHITECTURE.md.", "BASE DECISION: update OWNED_MODEL_TRAINING_STUDY.md + OWNED_MODEL_IMPLEMENTATION_PLAN.md to make Qwen3-VL the primary VLM base (was Qwen2.5-VL-7B); CLEAR the Gemma-4 'AMBER' (it is genuine Apache-2.0) but record Gemma-4 as bench-only (60s video cap, no timestamps) ? usable solely for native-audio cues. Fix the 'Qwen3-VL Nov 2025' date to Sept-Oct 2025 if it appears.", "STARTING POINT: execute M0.4 (productionize rally_seq_proto.py into a one-command ActionFormer/ASFormer train->eval->ship-gate harness over metrics.match_intervals) and M0.2 (n=3 -> n>=5 matches) FIRST. Both are \$0, local, no-cloud, no-DPA. The VLM/Gen-2 stays gated behind scaffolding + healthy corpus + explicit go.", "REFRAME THE NORTH-STAR IN DOCS: OWNED_MODEL_IMPLEMENTATION_PLAN.md still says

'on-device = the privacy moat / immediate' ? supersede that. On-device is a LATER upload-cost-saver; the immediate deliverable is a high-P/R extraction harness. Do not over-index the strategy on on-device now.", "BOOKKEEPING IS NO-REGRET, BUILD NOW: design the per-video event-index schema as a first-class data contract alongside the M0.1 corpus spine (it largely reuses the existing play_segments/events/candidate_segments tables). Reserve event/fault-tag taxonomy (e.g. service_fault) in M0.1 even if unpopulated. PUNT the NL parser/agent until quality clears ~0.583 LOVO.", "CONVERSATIONAL-QA ARCHITECTURE: lock the 'extract-once, event-store, structured-query (text-to-SQL + clip-cut)' contract early; reject end-to-end-VLM-per-turn; keep paid Gemini as inference/fallback only for open-ended edge questions.", "MULTISPORT: keep ONE shared head + ONE base; sequence per-sport data + clean trajectory detectors (badminton -> tennis/TT -> pickleball/padel). Do NOT cite RacketVision as evidence for unified temporal training (refuted ? it is ball-tracking only); treat single-model multisport generalization as an open hypothesis to validate, not a proven claim.", "WASB C3 PROVENANCE is a hard Gen-0 prerequisite that MULTIPLIES under multisport (each new sport needs a clean trajectory detector). Resolve via own-trained weights or a clean swap before any commercial ship; carry it as a tracked blocker, not an aside.", "LANGUAGE: no change ? Python everywhere; write down 'on-device runtime shell (Swift/Kotlin/C++/Rust)' as a deferred layer-local deployment item gated behind proven quality + measured upload-cost ROI. Impose the export-clean-ops constraint on the trained model.", "SEQUENCING DISCIPLINE HOLDS: this is a strategy recommendation, not a go-build order. The committed next platform slice remains the async job model (DEFERRED #16, single-tenant). Only the corpus/data/compliance/harness foundation (M0.1-M0.4) is safe to start, and even that needs the per-milestone owner greenlight already specified in the plan.", "licensing_watchlist": ["WASB shuttle/ball detector: code is MIT but the DISTRIBUTED checkpoint (wasb_badminton_best.pth.tar) is academic-data-trained with UNRESOLVED provenance (the C3 blocker). Propagates to anything trained on its features AND as a pseudo-label teacher. Must own-train weights or swap to a clean detector before commercial ship ? multiplies per sport.", "Gemini-silver TRAINING labels currently in training.label_candidates (over Gemini CSVs) VIOLATE the no-paid-LLM-training rule ? must be PURGED before they bake in (M0.3). Gemini stays eval/teacher-reference/fallback only.", "Public grounding-CoT datasets (e.g. TVG-R1's 56K) are Gemini-2.5-Pro-generated and therefore FORBIDDEN as a training source here ? the VLM path must originate its own CoT supervision (a real, binding cost).", "Gemma 3 (viral custom terms + PUP) ? REJECTED. Gemma 3n ('responsible commercial use' Gemma Terms that FOLLOW DERIVATIVES ? a fine-tune/distill stays bound) ? REJECTED, even cleaner reject than stated; do not adopt for on-device despite its edge/audio appeal.", "Llama 4 ? custom Community License with a 700M-MAU cap + 'Built with Llama' attribution ? NOT clean, REJECTED.", "Gemma 4 ? genuine standard Apache-2.0 (April 2026; clear the prior 'AMBER'); its separate Prohibited-Use doc could not be authoritatively confirmed 'non-binding' on the Apache grant ? have counsel confirm IF Gemma 4 ever becomes a serious contender (it is currently disqualified as primary extractor on the 60s/no-timestamp limits regardless).", "Qwen3-VL ? Apache-2.0 verified UNIFORM across 2B/4B/8B/235B + repo LICENSE; re-confirm the EXACT size you ship before release (every individual size/edition not opened). Note older Qwen2.5 forked to the custom Qwen License at 3B and 72B ? don't assume Qwen2.5 sizes are all Apache.", "DeepSeek-V4 (MIT) and InternVL3 (MIT + Apache-2.0 Qwen2.5 backbone) are commercial-clean; InternVL3's dual-license means tracking both obligations.", "unsloth: core is Apache-2.0 but the optional unsloth Studio UI is AGPL-3.0 ? keep Studio OUT of the proprietary product.", "Base-model PRETRAINING-data provenance is not publicly auditable for ANY open VLM (incl. Qwen3-VL). Your 'never train on paid-LLM outputs' guardrail is satisfied at the level of YOUR fine-tuning data; whether a vendor's base pretraining touched paid-LLM outputs is unknowable and governed by the base's license, not your pipeline ? flag for owner risk acceptance.", "Use sj-li/MS-TCN2 (MIT), NOT the yabufarha/ms-tcn repo (Commons Clause). MIT TAL heads (ActionFormer/ASFormer/TriDet) are clean.", "On-device convenience trap: 'Gemma 3 270M GGUF' is frequently cited for edge but is REJECTED by the guardrail ? keep on-device bases on the Qwen3-VL-4B Apache-2.0 lineage.", "open_questions": ["Does the single sport-agnostic model claim hold for TEMPORAL localization across sports? RacketVision (the cited evidence) was REFUTED as ball-tracking-only, so this is now an open hypothesis ? needs its own cross-sport eval once tennis/TT corpora exist.", "Does ActionFormer/ASFormer over the WASB feature stack beat the honest ~0.54 LOVO bar at SEGMENT level at n>=5 matches? The per-frame AUC (0.84-0.91) says likely, but threshold/operating-point transfer across videos is only partly solved at n=3.", "What is 'very high precision/recall' operationally? Define the ship-gate target IoU explicitly (e.g. F1@tIoU=0.5 for highlights, tighter for precise rally cuts) BEFORE the gate ? current best LOVO ~0.583 is far from 'very high' and the path there is more data + the segment-level head, not a base-model swap.", "Real VRAM + token cost of fine-tuning Qwen3-VL on actual 1080p/60fps clip->QA frame counts is UNVERIFIED for every framework (rollouts multiply video-token cost; ms-swift's documented multimodal-GRPO examples use 6-8 GPUs). Re-benchmark on a rented A100/L40S before budgeting Gen-2.", "Clean MIT/Apache/BSD trajectory detectors for tennis / table-tennis / pickleball / padel are NOT yet identified ? what fills the WASB role per

sport, and what is each one's checkpoint provenance?", "Realistic clip->QA label yield from ~128 rallies via augmentation + QA-multiplexing + own-teacher pseudo-labels ? does it reach the ~5k-20k band the VLM path needs, on amateur 60fps footage?", "Self-hosted-VLM \$/call + on-device latency/throughput of a quantized 4B on the 8GB 2070 Super ? both unmeasured and load-bearing for the owned-vs-Gemini crossover; benchmark before standing up any owned serving stack.", "Will a cloud provider (Modal primary; RunPod Secure-tier / Vertex asia-south1) sign a DPA with explicit no-train, India-configurable, for consented minors-excluded video?", "Is the MAU-cap rejection of larger Qwen/InternVL sizes over-conservative at solo-founder scale, and should 'Apache-2.0 only' be formally ratified to 'Apache-2.0/MIT' (the approved list already includes MIT)?", "sources": ["C:/Users/avidu/Projects/badminton-highlight-indexer/docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/PLATFORM_ARCHITECTURE.md", "C:/Users/avidu/Projects/badminton-highlight-indexer/backend/infrastructure/database.py (play_segments/events/candidate_segments + PRAGMA user_version ladder ? verified)", "C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/metrics.py (match_intervals scorer spine ? verified)", "C:/Users/avidu/Projects/badminton-highlight-indexer/backend/eval/rally_seq_proto.py (the de-risking per-frame prototype ? verified)", "https://github.com/QwenLM/Qwen3-VL + HF cards Qwen3-VL-2B/4B/8B/235B-Instruct (Apache-2.0 uniform; Text-Timestamp Alignment)", "https://arxiv.org/abs/2511.21631 (Qwen3-VL Technical Report)", "https://ai.google.dev/gemma/docs/core/model_card_4 (Gemma 4 Apache-2.0; 60s video cap; no timestamp output)", "https://huggingface.co/OpenGVLab/InternVL3-8B (MIT + Apache-2.0 backbone; multi-round video chat; no native timestamps)", "https://github.com/modelscope/ms-swift + https://swift.readthedocs.io/en/latest/BestPractices/GRPO-Multi-Modal-Training.html (Apache-2.0; native video+audio; multimodal GRPO custom-reward plugin)", "https://github.com/hiyouga/EasyR1 (image/text-only ? the gap that drops LLaMA-Factory)", "https://unsloth.ai/docs/get-started/reinforcement-learning-rl-guide/vision-reinforcement-learning-vlm-rl (Vision-RL image-only; SFT does video)", "https://arxiv.org/pdf/2503.13377 (Time-R1: fine-tuned VLM Charades-STA R@0.7=50.1 ? the boundary-precision ceiling)", "https://arxiv.org/pdf/2202.07925 (ActionFormer ~66.8% mAP THUMOS14 ? specialized-head boundary reference)", "https://arxiv.org/html/2512.01298 (TBT-Former 68.0% mAP, Dec-2025 ? heads still advancing)", "https://arxiv.org/html/2507.18100v1 (TVG-R1: ~13K SFT + ~18K RL recipe; public 56K CoT is Gemini-generated = forbidden)", "https://arxiv.org/html/2505.00254v3 (AVAS: event-knowledge-graph + RAG for long-video QA ? basis for query-over-index)", "https://arxiv.org/html/2412.09601v1 (TimeRefine: VLM-coarse-then-refine two-stage pattern)", "https://www.llama.com/llama4/license/ (Llama 4 700M-MAU cap ? rejected)", "https://ai.google.dev/gemma/terms (Gemma 3 / 3n custom terms follow derivatives ? rejected)", "https://onnxruntime.ai/docs/ (cross-language export boundary: train Python, run ONNX in C++/C#/Java/Rust)", "https://arxiv.org/html/2511.17045v2 (RacketVision ? VERIFIED as ball-TRACKING, NOT temporal; do not cite as temporal-generalization evidence)"]}]

RESEARCH:

```
[
  {
    "dimension": "BASE MODELS for a single, multisport, conversational video-QA model that also
does precise temporal localization (rally/highlight start-end)",
    "summary": "A commercially-clean base must satisfy four things at once: (1)
Apache-2.0/MIT/BSD weights with NO viral/NC/MAU restriction; (2) native video understanding of
long clips (full match, not a 60s window); (3) emits TIMESTAMPS for moment localization (the
rally/highlight start-end requirement); (4) holds multi-turn conversation about an uploaded
video (the north-star QA loop). Verified June 2026: Qwen3-VL is the only candidate that hits all
four cleanly. It ships Apache-2.0 UNIFORMLY across every size (2B/4B/8B/32B/30B-A3B/235B-A22B ?
no per-size license fork, unlike older Qwen generations), with a purpose-built \"Text-Timestamp
Alignment\" mechanism for second-level timestamp-grounded event localization, hours-long video
via 256K(?)1M) context, and native multi-turn multimodal chat. Gemma 4 (released April 2026) is a
genuine standard Apache-2.0 release (its Prohibited-Use Policy is non-binding guidance, not a
weights restriction ? no MAU cap, no commercial bar), and adds native audio (useful for racquet-
sport hit detection) ? but it caps video at ~60 seconds and documents NO temporal-
localization/timestamp capability, which directly disqualifies it as the PRIMARY extractor for
full-clip start-end localization. InternVL3 is license-clean (MIT project + Apache-2.0 Qwen2.5
LLM) and does multi-round video chat but has no documented native timestamp grounding. Gemma 3
is rejected (viral custom license per the existing guardrail); Gemma 3n is responsible-
commercial-license, not clean Apache-2.0, and edge-focused. Llama 4 / DeepSeek-V4 surface in
2026 roundups but Llama carries a custom community license with an MAU cap (not clean) and
neither documents racquet-grade temporal grounding. Recommendation: Qwen3-VL as the base family;
prototype the harness on the 4B/8B Instruct sizes (fit the 8GB 2070 Super for inference / small-
```

LoRA; rent cloud GPU for full fine-tunes and the larger sizes), keeping the option to scale to 30B-A3B/32B for the quality-first push and later distil down for the deferred on-device goal ? all within one Apache-2.0 family so no re-architecting.",

```
"options": [
  {
    "name": "Qwen3-VL (Alibaba) ? Instruct sizes 2B/4B/8B/32B + MoE 30B-A3B/235B-A22B",
    "what": "Multimodal (text+image+video) VLM series, Nov 2025, with a dedicated 'Text-
Timestamp Alignment' mechanism (beyond T-RoPE) for precise, second-level timestamp-grounded
event localization; 256K context expandable to 1M for hours-long video; native multi-turn
multimodal conversation; Thinking variants for reasoning. Active community fine-tuning + video-
grounding ecosystem (e.g. Qwen3-VL-Video-Grounding repo/Space).",
    "license": "Apache-2.0 (verified uniform across ALL sizes incl. the 235B ? no per-size
license fork)",
    "pros": [
      "Only candidate that natively emits timestamps for moment localization ? directly
matches the rally/highlight start-end requirement (Text-Timestamp Alignment, second-level
indexing)",
      "Apache-2.0 across EVERY size: clean for private proprietary commercial use; no MAU
cap, no NC clause; same license at 2B and 235B means no re-architecting when scaling quality up
or distilling down for the deferred on-device goal",
      "Handles hours-long video (256K?1M context) ? a full match clip, not a 60s window ? so
it can both localize and hold the contextual multi-turn QA the north-star feature needs",
      "Wide size ladder lets the 8GB 2070 Super run 4B/8B for inference and small LoRA
locally, while cloud-rented GPUs do full fine-tunes / larger sizes ? fits the solo-founder +
per-job cloud budget",
      "Sport-agnostic by construction (general VLM): one model can be fine-tuned across all
net-separated racquet sports, supporting the single sport-agnostic model goal without per-sport
bases",
      "Existing open fine-tuning + video-grounding tooling de-risks the training-harness
build (the immediate deliverable)"
    ],
    "cons": [
      "Base weights' training-data provenance is Alibaba's and not auditable; the Apache-2.0
license governs YOUR use, but if the owner extends the 'never train on paid-LLM' purity test to
the base's own pretraining, that cannot be fully verified (true of essentially every open VLM ?
treat as base-license-governed, not a YOUR-training-data issue)",
      "No native audio input (unlike Gemma 4) ? racquet-hit audio cues, if wanted, need a
separate audio branch or fusion",
      "Largest MoE sizes (235B) are impractical on the local box and pricey to fine-tune in
the cloud; the realistic working set is 4B?32B",
      "Quality on racquet-sport temporal localization is unproven for this corpus ? the de-
risking evidence (LOVO ~0.583) is from a separate per-frame prototype, not from this base; needs
its own fine-tune + eval"
    ],
    "fit": "strong"
  },
  {
    "name": "Gemma 4 (Google) ? E2B/E4B/12B/26B-A4B/31B",
    "what": "Google's April 2026 open-weight family; unified multimodal (text+image+video)
with NATIVE AUDIO on E2B/E4B/12B; multi-turn chat + system role; 128K?256K context. Released
specifically under standard Apache-2.0 (a deliberate break from Gemma 3's custom license).",
    "license": "Apache-2.0 (genuine standard Apache-2.0; the separate Prohibited-
Use/Intended-Use docs are non-binding policy guidance ? no MAU cap, no commercial-use bar)",
    "pros": [
      "Truly clean standard Apache-2.0 ? resolves the Gemma-3 viral-license rejection;
commercially usable in private proprietary software with no MAU cap",
      "Native AUDIO input (E2B/E4B/12B) ? directly useful for racquet-sport shot/hit
detection, a cue the badminton work already explored",
      "Strong size ladder incl. genuinely edge-friendly E2B/E4B for the eventual deferred
on-device compression goal",
      "Backed by Google with first-class tooling, quantization, and long-term support"
    ],
    "cons": [
      "DISQUALIFYING for the primary extractor: documents only ~60-second video input and NO
temporal-localization/timestamp capability ? cannot emit rally/highlight start-end over a full
```

```

match clip out of the box",
    "60s window also constrains the contextual multi-turn QA over a whole uploaded clip
(the north-star) without heavy external chunking/bookkeeping",
    "Temporal grounding would have to be taught from scratch via fine-tuning against a
base not designed to output timestamps ? higher risk/effort than Qwen3-VL's native support"
],
    "fit": "weak"
},
{
    "name": "InternVL3 (OpenGVLab) ? e.g. InternVL3-8B",
    "what": "Open multimodal series (April 2025) with strong video-understanding benchmarks
and documented multi-round video conversation; built on a Qwen2.5 LLM backbone. Successor lines
(InternVL3.5) add temporal-grounding gains via add-on methods.",
    "license": "MIT (project) + Apache-2.0 (Qwen2.5 LLM component) ? both clean for
proprietary commercial use",
    "pros": [
        "License-clean for commercial proprietary use (MIT + Apache-2.0)",
        "Strong video-understanding benchmarks and native multi-round video chat ? fits the
conversational QA loop",
        "Good size range; well-supported open ecosystem"
    ],
    "cons": [
        "No NATIVE timestamp/temporal-localization output documented in the base 3.x card ?
timestamp grounding relies on add-on methods (e.g. GroundVTS) or fine-tuning, weaker out-of-box
than Qwen3-VL for the start-end requirement",
        "Dual-license means tracking obligations of both the MIT project and the Apache-2.0
backbone (minor, but a compliance step)",
        "Effectively trails Qwen3-VL on the single most load-bearing requirement (native
timestamp emission)"
    ],
    "fit": "medium"
},
{
    "name": "Gemma 3 / Gemma 3n (Google, earlier)",
    "what": "Prior Gemma generation. Gemma 3 = multimodal but under the custom Gemma
license; Gemma 3n = edge/mobile multimodal (incl. audio) under a 'responsible commercial use'
license.",
    "license": "Gemma custom license (Gemma 3 ? viral/custom terms) / Gemma 'responsible
commercial use' (3n) ? NOT clean Apache-2.0/MIT/BSD",
    "pros": [
        "Gemma 3n is genuinely edge/on-device-oriented with audio ? relevant to the LATER
compression goal",
        "Mature tooling"
    ],
    "cons": [
        "FAILS the hard commercial-license guardrail: Gemma 3's custom/viral terms are
explicitly rejected by the owner; Gemma 3n's 'responsible commercial use' license is not a clean
Apache-2.0/MIT/BSD",
        "Superseded by Gemma 4's clean Apache-2.0 release anyway ? no reason to adopt the
restricted predecessor",
        "No advantage over Qwen3-VL on temporal localization"
    ],
    "fit": "weak"
},
{
    "name": "Llama 4 / DeepSeek-V4 (2026 general-LLM roundup mentions)",
    "what": "Large 2026 models surfaced in open-LLM roundups; Llama 4 Scout cited for very
long video context; DeepSeek-V4 variants cited as MIT with 1M context.",
    "license": "Llama = custom Llama Community License with an MAU cap (NOT clean);
DeepSeek-V4 = MIT (clean) per roundups",
    "pros": [
        "DeepSeek-V4 MIT licensing is commercially clean with very long context",
        "Llama 4 cited for long-video context windows"
    ],
    "cons": [

```

"Llama Community License carries an MAU-cap / acceptable-use restriction ? fails the 'reject >X-MAU caps if binding' and clean-license guardrail; not adoptable as the base",

"Neither documents racquet-grade NATIVE timestamp/temporal-localization for video moment localization ? primary requirement unmet",

"DeepSeek-V4's video-VLM + timestamp-grounding maturity is unverified vs Qwen3-VL's purpose-built support",

"Roundup-level sourcing only; not validated on the load-bearing temporal-grounding requirement"

```

    ],
    "fit": "weak"
  }
],
  "recommendation": "Adopt Qwen3-VL as the base-model family. It is the only verified June-2026 option that simultaneously satisfies all four load-bearing requirements: clean Apache-2.0 weights (uniform across every size ? no per-size license fork, no MAU cap, no NC clause), native long-video understanding (256K?1M context = full match clips, not a 60s window), purpose-built timestamp emission for moment localization (its 'Text-Timestamp Alignment' giving second-level event localization ? a direct match for rally/highlight start-end), and native multi-turn conversational video QA (the north-star loop). Build the training harness on the Instruct line: prototype on 4B/8B (runs on the 8GB 2070 Super for inference and small LoRA; full fine-tunes go to rented cloud GPU), reserve 30B-A3B/32B for the quality-first push, and keep the smaller sizes as the distillation target for the DEFERRED on-device goal ? all inside one Apache-2.0 family so scaling up or compressing down needs no re-architecting. Keep Gemma 4 explicitly on the bench for ONE reason: its native audio input (E2B/E4B/12B) is the cleanest path to racquet-hit audio cues if a fusion signal is later wanted ? but it cannot be the primary extractor (60s video cap, no timestamp output). Treat InternVL3 as the license-clean fallback if a Qwen-specific blocker emerges. Two scope notes for the harness, independent of base choice: (1) the conversational-QA north-star is best served by keeping EXTRACTION (the fine-tuned Qwen3-VL emitting timestamped rally/event spans) separate from the QA/RETRIEVAL layer (durable per-video bookkeeping + query routing over those spans) ? the model emits structured spans, the framework stores and answers over them; design that bookkeeping schema now (as the owner asks) but PUNT shipping conversational QA until rally/highlight quality is solid, since folding QA into the same fine-tune competes for the quality budget. (2) Base choice does NOT resolve the WASB checkpoint provenance blocker ? that detector sits below the VLM and stays an independent Gen-0 prerequisite.",
  "confidence": "high",
  "sources": [
    "https://github.com/QwenLM/Qwen3-VL",
    "https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct",
    "https://huggingface.co/Qwen/Qwen3-VL-2B-Instruct",
    "https://huggingface.co/Qwen/Qwen3-VL-235B-A22B-Instruct",
    "https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct",
    "https://arxiv.org/abs/2511.21631",
    "https://deepwiki.com/QwenLM/Qwen3-VL/5.4-video-understanding",
    "https://github.com/PRITHIVSAKTHIUR/Qwen3-VL-Video-Grounding",
    "https://ai.google.dev/gemma/docs/core",
    "https://ai.google.dev/gemma/docs/core/model_card_4",
    "https://ai.google.dev/gemma/apache_2",
    "https://ai.google.dev/gemma/prohibited_use_policy",
    "https://ai.google.dev/gemma/docs/gemma-3n",
    "https://en.wikipedia.org/wiki/Gemma_(language_model)",
    "https://huggingface.co/OpenGVLab/InternVL3-8B",
    "https://arxiv.org/pdf/2504.10479",
    "https://llm-stats.com/models/qwen3-vl-8b-instruct"
  ],
  "caveats": [
    "Per-size Apache-2.0 uniformity for Qwen3-VL was verified on the 2B, 4B(referenced), 8B, and 235B HF cards plus the GitHub repo footer ? all read apache-2.0; I did not individually open every single size/edition (e.g. 32B-Thinking, 30B-A3B FP8), but the consistency across the smallest and largest plus the repo-level license makes a per-size fork very unlikely. Re-confirm the exact size you ship before release.",
    "Gemma 4's Prohibited-Use Policy was assessed as non-binding policy guidance (no MAU cap, no commercial bar) from the policy page itself; if legal certainty is required, have counsel confirm whether Google treats it as a binding condition on the Apache-2.0 grant. This does not change the recommendation (Gemma 4 is disqualified on the 60s/no-timestamp technical limits

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regardless).",

"The Gemma 4 ~60-second video cap and 'no temporal localization' come from the official model card/overview as of this search; Google could extend these ? recheck if Gemma 4 becomes a serious contender. Not load-bearing for the recommendation.",

"Base-model PRETRAINING data provenance (for ALL candidates, incl. Qwen3-VL) is not publicly auditable. The owner's 'never train on paid-LLM outputs' guardrail is satisfied at the level of YOUR fine-tuning data; whether a vendor's base pretraining touched paid-LLM outputs is unknowable and is governed by the base's license, not your training pipeline. Flag for owner's risk acceptance.",

"Qwen3-VL's quality on THIS racquet-sport temporal-localization task is unproven ? the cited de-risking evidence (LOVO ~0.583, AUC 0.83?0.92) is from a separate per-frame prototype, not this base. The recommendation is on FIT/license/capability, not a benchmarked accuracy claim; the harness must run its own fine-tune + cross-video eval.",

"2026 'best open LLM' roundups (Llama 4, DeepSeek-V4) are secondary/blog sources and were not individually license-verified at the model-card level; they are included only to show they were considered and ruled out on the temporal-grounding and (for Llama) MAU-cap grounds, not as adoptable candidates."

]

},

{

"dimension": "Fine-tuning frameworks for video VLMs (SFT + RFT/GRPO for temporal grounding + conversational video-QA)",

"summary": "FIRST, THE ORTHOGONALITY THE OWNER ASKED ME TO CLARIFY: a *framework* (ms-swift, LLaMA-Factory, axolotl, unsloth, TRL) and a *base model* (Qwen3-VL, Qwen2.5-VL, Qwen3-Omni, Gemma-4) are independent choices. The framework is the \"forge\" that runs SFT/GRPO on a GPU and emits LoRA/full weights; the base is the metal you forge. You pick ONE framework and feed it ANY Apache/MIT base, and a different base later (Qwen->Gemma) is a config swap, not a re-platforming. So this dimension does NOT pick the model; it picks the trainer. ALL FIVE leading frameworks are permissively licensed (Apache-2.0; unsloth's only AGPL piece is the optional Studio UI, trivially avoided), so the hard commercial guardrail does not eliminate any of them at the framework layer; license cleanliness is decided by the BASE WEIGHTS + DATA, not the trainer. The differentiator is capability on the four hard requirements: (1) NATIVE video training, (2) NATIVE audio training, (3) timestamp/temporal-grounding outputs, (4) RFT/GRPO with a CUSTOM reward ON VIDEO inputs, runnable on a rented A100/L40S. On those four jointly, exactly one framework clears all four in a single stack: ms-swift. It is the only framework that advertises and documents native video+audio training AND multimodal GRPO with a custom external-reward plugin executed over video rollouts (via vLLM), plus the fps/num_frames video-sampling controls you need to train timestamped grounding. The others each fail at least one hard requirement: LLaMA-Factory does native video+audio SFT but DELEGATES GRPO to a separate repo (EasyR1) that is image-only; unsloth's much-praised Vision-RL/GRPO is image-only (its value is the 8GB-box SFT, which matters LATER for compression, not now); TRL's GRPO-VLM is image-only and you would hand-build the video reward loop; axolotl explicitly labels video \"not well tested.\" NOTE ON TEMPORAL GROUNDING: emitting timestamps is primarily a BASE-MODEL + dataset-format property (Qwen2.5/3-VL align MRoPE position IDs to wall-clock time; you train on a JSON messages format with second/HMS timestamps), so the framework's job is just to expose video-frame controls and not corrupt the time encoding, which ms-swift does and is the most-documented for it. RECOMMENDATION: adopt ms-swift as the single training/RFT forge now; keep unsloth in the back pocket as the 8GB-local SFT/compression tool for the deferred on-device goal. This aligns with, and now hardens, the lean already recorded in docs/PLATFORM_ARCHITECTURE.md (which favored ms-swift but predates the RFT-custom-reward and audio requirements that make the gap decisive).",

"options": [

{

"name": "ms-swift (ModelScope SWIFT)",

"what": "Self-managed CLI/Python fine-tuning + RL + deploy framework, 600+ LLMs / 300+ MLLMs. The ONLY framework that natively trains video AND audio (Qwen3-VL, Qwen3-Omni, Gemma-4, InternVL3.5) and runs multimodal GRPO with a custom external-reward plugin over video rollouts (vLLM-accelerated). Exposes fps/num_frames video sampling; JSON messages dataset format carries timestamps for temporal grounding. Has an eval backend (EvalScope) and args.json reproducibility, fitting the agent-runnable report-card harness goal.",

"license": "Apache-2.0",

"pros": [

"ONLY framework meeting all four hard requirements in one stack: native video + native audio + timestamp/grounding controls + GRPO/RFT with a CUSTOM reward on VIDEO inputs (documented Qwen2.5-VL GRPO example uses --external_plugins + --reward_funcs + --use_vllm)",

```

    "Apache-2.0, clean for private/proprietary/commercial use at the framework layer; no
    MAU cap or NC clause",
    "Supports the exact bases on your approved list (Qwen3-VL, Qwen2.5-VL, Qwen3-Omni for
    audio, Gemma-4) so the framework decision does not constrain the later base choice",
    "Any model size incl. cheap-to-serve 7B/4B; runs on rented A100/L40S (your
    Modal/RunPod plan); multimodal GRPO examples are 6-8 GPU but SFT/LoRA fits a single 24-48GB
    card",
    "Custom reward plugin is the seam to reuse your existing metrics.match_intervals
    temporal-IoU scorer as the RFT reward (the study Gen-2 plan): single scorer spine, no metric
    drift",
    "Reproducibility (args.json, full_determinism) + EvalScope eval backend map onto the
    nanochat-style one-command report-card harness you want",
    "Active, large project (AAAI 2025; v4.0 Mar-2026), broadest model coverage of any
    option"
  ],
  "cons": [
    "Heaviest/most-configurable surface area, many flags; steeper first-run than unsloth
    notebooks",
    "Multimodal GRPO is VRAM-hungry (multiple video rollouts/sample); real video-token
    VRAM on 1080p/60fps clips is UNVERIFIED (same open risk flagged in the training study); re-
    benchmark before budgeting",
    "Audio-modality (Omni) training is less battle-tested than its video/image path;
    verify the Qwen3-Omni recipe before relying on audio cues",
    "ModelScope-centric defaults (some model pulls assume ModelScope hub); point it at HF
    mirrors for clean provenance"
  ],
  "fit": "strong"
},
{
  "name": "LLaMA-Factory (+ EasyR1 for RL)",
  "what": "Popular, well-documented self-managed SFT/DPO framework for 100+ LLMs/VLMs with
  native video (LLaVA-NeXT-Video) and audio (Qwen2-Audio, Qwen2.5-Omni) SFT. GRPO is NOT in-core;
  it is delegated to the sibling repo EasyR1 (a veRL fork), which is image/text-only.",
  "license": "Apache-2.0 (both LLaMA-Factory and EasyR1)",
  "pros": [
    "Apache-2.0, very approachable, best-documented onboarding; native video + audio SFT
    in-core",
    "Strong if your near-term need is SFT-only cold-start (Gen-2 SFT cold-start step)
    before any RL",
    "Large community, many tutorials, lowest ramp for a solo founder doing supervised
    runs"
  ],
  "cons": [
    "DECISIVE GAP: GRPO/RFT is delegated to EasyR1, which is IMAGE-ONLY (no video, no
    audio, no confirmed custom-reward-on-video path). So your RFT-with-temporal-IoU-reward-on-video
    requirement is NOT met in one stack; you would run two frameworks and EasyR1 still cannot take
    video",
    "Splitting SFT (LLaMA-Factory) and RL (EasyR1) doubles the harness surface and breaks
    the single-command report-card goal",
    "Video SFT works but the framework is not the strongest-documented for fine-grained
    timestamp grounding vs ms-swift Qwen-VL best-practices"
  ],
  "fit": "medium"
},
{
  "name": "Unsloth",
  "what": "Single-GPU-optimized framework (70-90% less VRAM, 1.5-2x faster) with Qwen3-VL
  SFT and a celebrated Vision-RL (GRPO/GSP0) path. Trains Qwen3-VL-8B GRPO on a free Colab T4. The
  RL path is IMAGE-input only.",
  "license": "Apache-2.0 core (optional Unsloth Studio UI is AGPL-3.0, avoidable; do not
  ship the Studio component)",
  "pros": [
    "Best-in-class for the 8GB RTX 2070 Super: can SFT/LoRA a 7B/8B VLM locally where
    others OOM, directly serving the DEFERRED on-device/compression goal",
    "Apache-2.0 core; vision GRPO/GSP0 supported (image), 256K context",

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    "Cheapest first runs / fastest local iteration; great for local SFT cold-start and
    later quantization/compression"
  ],
  "cons": [
    "Vision-RL is IMAGE-ONLY, does not satisfy the RFT-with-custom-reward-ON-VIDEO
    requirement central to the quality goal",
    "Single-GPU focus; not the multi-GPU rented-A100/L40S RFT engine for video",
    "Tightly coupled to its own patched kernels/model classes, less of a general swap-any-
    base + any-reward forge than ms-swift",
    "Keep the AGPL Studio UI out of the proprietary product to stay license-clean"
  ],
  "fit": "medium"
},
{
  "name": "HuggingFace TRL (+ transformers)",
  "what": "The foundational RL/post-training library (GRPO, DPO, PPO, RL00) that most
  other frameworks build on. Multiple weighted custom reward functions are first-class. Its
  documented GRPO-VLM path is image+text.",
  "license": "Apache-2.0",
  "pros": [
    "Apache-2.0, maximally portable, the de-facto substrate; owning the recipe here is the
    ultimate anti-lock-in hedge",
    "Cleanest, most flexible CUSTOM reward API (list of reward fns, reward_weights), ideal
    abstraction for your temporal-IoU reward",
    "Tight transformers integration; any HF Apache/MIT base loads directly with clean
    provenance"
  ],
  "cons": [
    "GRPO-VLM is documented/tested for IMAGE inputs; video rollout plumbing (frame
    sampling, video reward batching) is DIY, you become the framework author",
    "No batteries-included native video/audio dataset pipeline or fps/num_frames
    ergonomics, slowest path to a working video RFT loop",
    "Lower-level: more glue code for a solo founder vs ms-swift ready video-GRPO examples"
  ],
  "fit": "medium"
},
{
  "name": "Axolotl",
  "what": "Production-oriented YAML-config SFT/DPO/GRPO framework with a BETA multimodal
  path (Qwen2-VL, Pixtral, SmolVLM2 video, Voxtral/Gemma-4 audio).",
  "license": "Apache-2.0",
  "pros": [
    "Apache-2.0; clean reproducible YAML configs, good production ergonomics for SFT",
    "Has audio (Voxtral, Gemma-4) and some video models; LoRA/QLoRA efficient"
  ],
  "cons": [
    "DECISIVE GAP: video support is explicitly documented as not well tested at the
    moment, the wrong bet for a VIDEO-FIRST very-high-quality goal",
    "Multimodal is framework-level BETA; custom-reward-on-video GRPO not demonstrated",
    "Strengths (text SFT ergonomics) are not where this project hard requirements live"
  ],
  "fit": "weak"
}
],
"recommendation": "Adopt ms-swift (Apache-2.0) as the single SFT+RFT forge now. It is the
only framework that meets all four hard requirements jointly: native video, native audio
(Qwen3-Omni/Gemma-4), timestamp/grounding via fps/num_frames + the JSON messages format, and
GRPO/RFT with a CUSTOM external-reward plugin executed over VIDEO rollouts (vLLM); and it runs
on the rented A100/L40S you already plan to use (Modal/RunPod). Its custom-reward plugin is the
exact seam to wire your existing metrics.match_intervals temporal-IoU scorer in as the RFT
reward (the Gen-2 plan in OWNED_MODEL_TRAINING_STUDY.md), keeping one scorer spine across
heuristic-labeling, eval, and RL. The competitors each fail a hard requirement: LLaMA-Factory
does native video+audio SFT but offloads GRPO to EasyR1, which is image-only (so the video-RFT
goal needs a two-framework Frankenstein that still cannot take video); unsloth's Vision-RL and
TRL's GRPO-VLM are both image-only; axolotl's video path is self-described as not well tested.

```

CLARIFY FOR THE OWNER (the orthogonality you asked me to state): this decision is independent of the base-model choice; ms-swift trains Qwen3-VL, Qwen2.5-VL, Qwen3-Omni (audio), and Gemma-4 alike, so which-framework and which-base are decoupled and the base can change later without re-platforming. Because every leading framework is permissive, the commercial-license guardrail is NOT decided here; it is decided by the base WEIGHTS provenance and the training DATA (your WASB-checkpoint C3 blocker and the no-paid-LLM-labels rule live at the data/weights layer, not the trainer). Keep unsloth as the secondary, 8GB-local tool for the DEFERRED on-device/compression goal (it is the best fit for the RTX 2070 Super), and keep TRL in view as the portability hedge since ms-swift, unsloth, and EasyR1 all sit on TRL/veRL-style primitives anyway; you own the data + recipe + exported weights, so the forge stays swappable. ON CONVERSATIONAL VIDEO-QA / BOOKKEEPING: the framework choice does not force the embark-now vs punt decision; the per-video state (rally inventory, shot counts, fault tags, derived clips) is an APPLICATION/RETRIEVAL layer that NO trainer holds; ms-swift just lets you later add a QA-style SFT task over the same corpus when quality is solid. So pick the forge now, build the bookkeeping store from the start as the brief requires, and keep conversational-QA as an SFT task you can add to ms-swift later rather than a framework prerequisite.",

```
"confidence": "high",
"sources": [
  "https://github.com/modelscope/ms-swift",
  "https://swift.readthedocs.io/en/latest/BestPractices/GRPO-Multi-Modal-Training.html",
  "https://qwen.readthedocs.io/en/latest/training/ms_swift.html",
  "https://github.com/modelscope/ms-swift/issues/7326",
  "https://github.com/hiyouga/LLaMA-Factory/blob/main/README.md",
  "https://github.com/hiyouga/EasyR1",
  "https://docs.axolotl.ai/docs/multimodal.html",
  "https://www.spheron.network/blog/axolotl-vs-unsloth-vs-torch-tune/",
  "https://unsloth.ai/docs/get-started/reinforcement-learning-rl-guide/vision-reinforcement-learning-vlm-rl",
  "https://unsloth.ai/docs/models/qwen3-how-to-run-and-fine-tune/qwen3-vl-how-to-run-and-fine-tune",
  "https://github.com/unslothai/unsloth",
  "https://github.com/huggingface/trl/blob/main/docs/source/grpo_trainer.md",
  "https://huggingface.co/learn/cookbook/fine_tuning_vlm_grpo_trl",
  "https://pyimagesearch.com/2025/06/16/video-understanding-and-grounding-with-qwen-2-5/",
  "https://arxiv.org/pdf/2505.20715",
  "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/PLATFORM_ARCHITECTURE.md",
  "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md"
],
"caveats": [
  "VRAM/token cost of multimodal GRPO on real 1080p/60fps clips is UNVERIFIED for every framework (rollouts multiply video-token cost); the documented ms-swift multimodal GRPO examples use 6-8 GPUs; re-benchmark on a rented A100/L40S before budgeting. This is the same open risk already flagged in OWNED_MODEL_TRAINING_STUDY.md, not new.",
  "Framework license is clean across the board, but that does NOT clear the COMMERCIAL guardrail; license cleanliness is decided by the BASE WEIGHTS provenance (e.g. the unresolved WASB checkpoint C3 blocker) and the TRAINING DATA (no paid-LLM labels; minors excluded). Do not let framework-is-Apache-2.0 imply the pipeline is clean.",
  "ms-swift audio (Qwen3-Omni / Gemma-4) training path is less battle-tested than its image/video path; verify the Omni recipe end-to-end before committing to audio as a training cue. Audio was NOT adopted for fusion in prior work (PR #35, audio-E1 findings), so treat audio-in-training as exploratory.",
  "Temporal-grounding/timestamp emission is mostly a BASE-MODEL + dataset-format property (Qwen MROPE time alignment), not a framework feature; choosing ms-swift gives you the controls (fps/num_frames, timestamped messages) but the grounding QUALITY depends on the base and the data format, which must be validated separately.",
  "Web specifics (version numbers, exact model lists, GPU counts in examples) are 2026 snapshots and move fast; re-confirm the live ms-swift docs/version at implementation time. Implementation remains gated on owner sign-off + PLATFORM scaffolding per the existing sequencing directive; this is a framework SELECTION, not a go-build order.",
  "EasyR1 being image-only is the load-bearing reason LLaMA-Factory drops to medium; if EasyR1 (or LLaMA-Factory in-core) adds video GRPO later, re-evaluate; LLaMA-Factory SFT ergonomics are otherwise excellent for the cold-start step."
]
},
```

```

{
  "dimension": "Model strategy for high-P/R rally+highlight temporal extraction (multisport,
single model): fine-tuned VLM vs specialized TAL head vs both",
  "summary": "The evidence says BOTH, in a deliberate sequence ? and it strongly validates the
project's existing 3-generation blueprint (TAS head now ? VLM later) while sharpening the
rationale. The core empirical fact: a fine-tuned VLM does NOT reach \"very high\"
precision/recall at STRICT temporal boundaries. The strongest 2025-26 RL-post-trained video VLM
(Time-R1 on Qwen2.5-VL-7B) reaches only R@0.7=50.1% mIoU=60.9% on Charades-STA *fine-tuned*; the
same base Qwen2.5-VL-7B off-the-shelf is R@0.7=24.3%. Specialized TAL heads remain ahead on
boundary precision (ActionFormer ~66.8% mAP on the harder multi-instance THUMOS14; Dec-2025 TBT-
Former 68.0%). The literature consensus is explicit: VLMs \"capture coarse temporal regions
rather than accurate event boundaries\" and \"struggle with absolute fine-grained temporal
grounding\" ? which is exactly the rally start/end accuracy this product sells. So: the TAL/own-
feature head is the right vehicle for the *quality-first* extraction goal NOW (cheap, $0 local
on the 2070, already de-risked to AUC 0.84-0.91 / LOVO 0.615 at n=3), and the VLM is the right
vehicle for the *conversational-QA north star* LATER ? but evidence says the QA layer should be
RAG/event-index over the head's structured output, NOT a single end-to-end VLM that does both
detection and conversation. A single sport-agnostic model is feasible and even beneficial:
RacketVision (AAAI'26, badminton+tennis+TT) shows unified multi-sport training BOOSTS
generalization +15-19% mAP, at a modest cost to per-sport pinpoint precision ? directly
answering the multisport tension. Recommendation: BOTH, sequenced ? own TAL head as the quality-
first extractor + teacher/label-amplifier and the honest baseline; VLM as the deferred
conversational target fed by the head's event index. Embark on the bookkeeping/event-index
framework NOW (it's no-regret and the QA layer needs it regardless); PUNT the end-to-end
conversational VLM until extraction quality is solid and the corpus is large.",
  "options": [
    {
      "name": "Option A ? Specialized TAL head over own feature stack
(ActionFormer/ASFormer/TriDet), per-sport-agnostic, single shared head",
      "what": "A tiny temporal-action-localization/segmentation head (ActionFormer 1:1
rally=instance, or ASFormer/MS-TCN2 for per-frame TAS) trained on a hand-built per-frame feature
stack derived from a ball/shuttle trajectory detector (WASB-class) plus motion/audio/pose cues.
One shared, sport-agnostic head; sport identity is just another input feature or a per-sport
calibration, not a separate model. This is the project's de-risked Gen-0/Gen-1 path.",
      "license": "MIT (ActionFormer, ASFormer, TriDet, MS-TCN2 ? use sj-li/MS-TCN2 not the
Commons-Clause yabufarha repo). Trajectory detector code MIT (WASB) but the distributed
checkpoint provenance is UNRESOLVED (C3 blocker) ? own-weights or a clean alternative needed.",
      "pros": [
        "Highest boundary precision per dollar ? specialized heads still lead VLMs at strict
IoU (ActionFormer ~66.8% mAP THUMOS14 multi-instance; TBT-Former 68.0% Dec-2025). Boundary
accuracy IS the product.",
        "Already de-risked on THIS corpus: learned per-frame prototype AUC 0.84-0.91 held-out,
LOVO mean 0.615 at n=3, overtaking the heuristic and decoupling from the multi-court confound.",
        "Trains in minutes, ~4.5GB VRAM, $0 on the local 2070 Super ? fits the solo-
founder/single-GPU budget; no cloud DPA needed for training.",
        "Tiny (~1M params) ? trivially on-device later, satisfying the deferred compression
goal without re-architecting.",
        "Sport-agnostic single head is validated: RacketVision shows unified multi-sport
training improves generalization +15-19% mAP. The feature stack (trajectory
visibility/velocity/net-crossing) is a racket-sport invariant.",
        "Doubles as the label-amplifier/teacher (confidence-gated pseudo-labels) and the
honest baseline the VLM must beat ? pure upside even if the VLM later wins."
      ],
      "cons": [
        "Cannot do conversation / open-ended QA ? it only emits [start,end,label]; the north-
star QA feature must live in a separate retrieval/LLM layer on top.",
        "Generalization to tennis/TT/padel is asserted-but-unproven on THIS stack: the corpus
and WASB features are badminton-only today; needs per-sport trajectory detectors (clean-license
ones not yet identified for all 5 sports) + per-sport labeled data.",
        "Depends on a clean trajectory detector; WASB checkpoint provenance (C3) is an open
commercial-cleanliness blocker that must be resolved (own-train weights or swap, e.g. MS-
TrackNetV3-class) before commercial ship.",
        "Feature engineering is per-sport-fragile at the edges (net height, ball speed regimes
differ) even if the schema generalizes ? some per-sport calibration is likely."
      ]
    }
  ]
}

```

```

    "fit": "strong"
  },
  {
    "name": "Option B ? Fine-tuned single end-to-end VLM (Qwen3-VL-4B/Qwen2.5-VL-7B) doing
detection + segmentation + conversational QA",
    "what": "One Apache-2.0 video VLM fine-tuned to emit rally [start,end] timestamps AND
answer contextual questions, via SFT cold-start then GRPO/RFT with a temporal-IoU-vs-golden
reward. The owner's 'hopefully one conversational model' end-state.",
    "license": "Apache-2.0 bases exist and are clean: Qwen3-VL-4B/8B, Qwen2.5-VL-7B/32B,
Gemma-4 E4B/12B-Unified (keep AMBER until raw HF LICENSE verified). BUT the standard public
grounding-CoT training data (e.g. TVG-R1's 56K) is Gemini-2.5-Pro-generated ? FORBIDDEN here;
you must build your own CoT labels, which is the binding cost.",
    "pros": [
      "Only path to the conversational-QA north star within a single model ? answers 'show
me the longest rally', 'all the service faults', natural-language slicing.",
      "Generalizes across sports/play-types with far less per-sport feature engineering ?
semantic priors transfer; one model, many sports, fewer bespoke detectors.",
      "Rapidly improving: 2025-26 RL post-training (Time-R1, MUSEG, TimeRFT) lifted
Qwen2.5-VL-7B from R@0.7=24.3% (off-the-shelf) to 50.1% fine-tuned; Qwen3-VL adds explicit text-
timestamp alignment for second-level indexing.",
      "Same labeled corpus feeds it via the clip2QA transcoder ? no re-labeling, consistent
with the moat design."
    ],
    "cons": [
      "Does NOT hit 'very high' P/R at strict boundaries: best fine-tuned VLM is R@0.7?50%
mIoU?61% on Charades-STA ? half of moments still misaligned at IoU 0.7. The literature is
explicit VLMs 'capture coarse regions, not accurate boundaries.' This is a direct miss on the
quality-first goal.",
      "Forbidden-teacher penalty is real and binding: the proven data recipe (~13K SFT +
~18K RL CoT) leans on paid-LLM-generated CoT labels, which this project bans ? you must
originate that supervision yourself, raising the bar substantially.",
      "Cannot train locally: QLoRA of a 7B video VLM needs ~24GB+; requires rented cloud
(Modal/RunPod) + signed DPA/no-train for consented footage. GRPO multiplies cost (many rollouts
over long visual contexts).",
      "On-device path is heavier (4B quantized on 8GB Turing is unbenchmarked) ? defers, not
deletes, the compression problem.",
      "Holding per-video QA STATE (rally inventory, shot counts, fault tags) is NOT
something the model itself does reliably ? even the single-model dream still needs external
bookkeeping, so it doesn't eliminate the framework requirement."
    ],
    "fit": "weak"
  },
  {
    "name": "Option C (RECOMMENDED) ? BOTH, sequenced: own TAL head as quality-first
extractor + teacher/baseline; VLM deferred as the conversational target fed by a RAG/event-index
over the head's output",
    "what": "Phase 1: ship the sport-agnostic TAL head as the high-P/R rally/highlight
extractor (the de-risked Gen-0/1). It writes a durable per-video EVENT INDEX (rally inventory,
[start,end], shot counts, ending_reason/fault tags, derived-clip refs). Phase 2 conversational
QA = an LLM/agent doing RETRIEVAL over that index (AVAS-style event knowledge graph + RAG),
composing answers/clips deterministically ? NOT re-deriving events. The head also serves as a
confidence-gated pseudo-label teacher and the honest baseline. A fine-tuned VLM is a LATER,
optional upgrade for the grounding/QA stage once the corpus is large and quality is solid ? and
even then likely as a coarse-proposer refined by the head (the TimeRefine/perception-decoder
two-stage pattern), not a wholesale replacement.",
    "license": "All-clean: MIT TAL heads + Apache-2.0 LLM for the QA/retrieval layer (Qwen-
class) + own/clean trajectory weights. No paid-LLM in any TRAINING path; paid Gemini stays
inference/fallback only. The QA-layer LLM does inference-time reasoning over YOUR structured
index (allowed) ? it is not trained on paid-LLM outputs.",
    "pros": [
      "Hits the quality-first goal NOW with the boundary-accurate head, while keeping the
conversational north star reachable ? no false choice between them.",
      "RAG-over-event-index is the 2025-26 best practice for long-video contextual QA (AVAS
event knowledge graph, MemoryCard): more accurate, cheaper, and auditable than end-to-end VLM
QA, and it's deterministic for 'make a clip of all X'."
    ]
  }
}

```

"Satisfies 'bookkeeping from the start': the event index IS the per-video state the QA loop needs ? build it now as a no-regret framework artifact even if QA ships later.",

"Decouples risk: extraction quality and conversational capability evolve independently; you never block shipping a good extractor on an unproven VLM, and you can A/B a VLM proposer behind the same scorer spine.",

"Cheapest commercially-clean route: \$0 local training for the head; the QA LLM is inference-only over your data; the VLM cloud-training spend is deferred until justified by QA volume and a \$/call crossover.",

"Same one corpus + one scorer (match_intervals / tIoU) feeds head training, TAL eval, pseudo-labeling, AND the eventual VLM reward ? zero metric drift, the moat stays the dataset+harness."

],

"cons": [

"Two systems to build and integrate (extractor + retrieval/QA layer) rather than one model ? more surface area, a clear data contract (the event-index schema) is required up front.",

"The conversational answers are only as good as the head's event tags ? richer semantics (e.g. 'service fault') need explicit event types the head/labels must support, adding label taxonomy work.",

"Defers the 'single conversational model' aesthetic the owner hopes for; if a single-model VLM later proves it can do both at high P/R, some of the retrieval scaffolding is superseded (but the index/corpus is reused, so low regret).",

"Still inherits Option A's open blockers: WASB C3 provenance and multisport feature/detector generalization must be resolved."

],

"fit": "strong"

}

],

"recommendation": "Adopt Option C ? BOTH, strictly sequenced ? which is essentially a sharpened version of the project's existing 3-generation blueprint with two course-corrections justified by the fresh evidence. Concretely: (1) START NOW with the specialized, sport-agnostic TAL head (ActionFormer/ASFormer-class, MIT) over the own feature stack as the QUALITY-FIRST rally/highlight extractor ? it is the only path that credibly reaches very-high P/R at strict boundaries on a \$0 local budget, and it's already de-risked on this corpus (AUC 0.84-0.91, LOVO 0.615 at n=3). It triples as extractor, confidence-gated label-amplifier/teacher, and the honest baseline the VLM must beat. (2) DESIGN THE PER-VIDEO EVENT-INDEX/BOOKKEEPING FRAMEWORK NOW ? this is no-regret: the conversational-QA loop needs durable per-video state (rally inventory, [start,end], shot counts, fault/event tags, derived-clip refs) regardless of whether QA is a VLM or a retrieval layer. Make the index schema a first-class data contract alongside the labeled-corpus spine (M0.1). (3) PUNT the end-to-end conversational VLM. Evidence is decisive that a fine-tuned VLM does not hit very-high P/R at strict temporal boundaries (best-in-class R@0.7?50%), so folding detection+QA into one model would REGRESS the core quality metric the product sells. Build conversational QA, WHEN it ships, as an LLM/agent doing RAG over the head's event index (AVAS-style event-knowledge-graph) ? the 2025-26 best practice for long-video contextual QA, and deterministic for 'make a clip of all service faults.' (4) Keep a fine-tuned VLM as a LATER, OPTIONAL upgrade for the grounding/QA stage once the corpus is large (the proven recipe is ~13K SFT + ~18K RL examples ? and note the public grounding-CoT data is Gemini-generated and thus forbidden here, so you must originate that supervision, which raises the bar). If/when adopted, prefer the two-stage pattern (VLM coarse-proposes, the head/decoder refines boundaries ? TimeRefine/perception-decoder), not wholesale replacement. (5) On MULTISPORT: a single sport-agnostic model is feasible and beneficial ? RacketVision (AAAI'26) shows unified badminton+tennis+TT training improves generalization +15-19% mAP at a modest pinpoint-precision cost; keep one shared head with sport as a feature/calibration. The binding multisport blocker is not the model ? it's clean per-sport trajectory detectors + per-sport labeled data, and the WASB checkpoint C3 provenance, all of which must be resolved before commercial ship. Net: do NOT start with a VLM; do NOT skip the head; build the bookkeeping now; treat the VLM as a deferred QA-stage option, not the quality-first extractor.",

"confidence": "high",

"sources": [

"https://arxiv.org/html/2508.10922v1 ? A Survey on Video Temporal Grounding with Multimodal LLMs (2025): VTG-MLLMs 'gradually surpassing' fine-tuned methods but 'struggle to capture fine-grained temporal dependencies' and 'precise temporal boundary prediction'; zero-shot Time-R1 R@0.5=60.8 R@0.7=35.3 on Charades-STA",

"https://arxiv.org/pdf/2503.13377 ? Time-R1 (Qwen2.5-VL-7B, GRPO RL): FINE-TUNED Charades-STA R@0.3=82.8 R@0.5=72.2 R@0.7=50.1 mIoU=60.9 ? best-class VLM still only ~50% at strict

IoU=0.7",
 "https://arxiv.org/pdf/2502.13923 ? Qwen2.5-VL Technical Report: 7B off-the-shelf Charades-STA R@0.3=67.9 R@0.5=50.3 R@0.7=24.3 mIoU=43.6 (the project's candidate base, pre-fine-tune)",
 "https://arxiv.org/pdf/2202.07925 ? ActionFormer (MIT): 66.8% avg mAP THUMOS14, 36.6 avg mAP ActivityNet 1.3 ? specialized TAL boundary-precision reference",
 "https://arxiv.org/html/2512.01298 ? TBT-Former (Dec 2025): 68.0% avg mAP THUMOS14, new SOTA, surpassing ActionFormer ? specialized heads still advancing",
 "https://arxiv.org/html/2511.17045v2 ? RacketVision (AAAI 2026, badminton+tennis+TT, 1,672 clips): unified multi-sport MS-TrackNetV3 beats single-sport +19.2% mAP tennis / +14.6% badminton, with a modest pinpoint-precision tradeoff ? validates single sport-agnostic model",
 "https://arxiv.org/html/2412.09601v1 ? TimeRefine: VLM coarse-predict-then-refine-offsets two-stage; evidence the boundary-refinement/decoder pattern is needed for fine VLM localization",
 "https://arxiv.org/html/2507.18100v1 ? TVG-R1 (EMNLP 2025 Industry): proven VLM grounding recipe = ~13K SFT cold-start + ~18K RL; public 56K CoT labels generated by Gemini-2.5-Pro (forbidden as a training source here)",
 "https://arxiv.org/html/2505.00254v3 ? AVAS: event-knowledge-graph index + RAG agentic reasoning for long-video QA ? best-practice basis for the conversational layer over an event index rather than end-to-end VLM",
 "https://arxiv.org/abs/2511.21631 ? Qwen3-VL Technical Report: explicit text-timestamp alignment for second-level video indexing (improved grounding base, Apache-2.0)"
],
 "caveats": [
 "Benchmark transfer caveat: Charades-STA/THUMOS14/ActivityNet are open-domain. They establish the VLM-vs-head BOUNDARY-PRECISION ceiling pattern, but absolute numbers won't map 1:1 to amateur 1080p/60fps racket footage ? rallies are more periodic/visually distinctive than generic captioned moments, so the head's relative advantage may be even larger here (favoring the recommendation), but this is inference, not measured.",
 "Multisport generalization of THIS feature stack is unproven: the de-risking corpus and WASB features are badminton-only. RacketVision validates the SINGLE-MODEL concept (and a clean MS-TrackNetV3-class detector), but tennis/TT/pickleball/padel each need clean-license trajectory detectors + labeled data not yet in hand. Pickleball/padel have far less public dataset/detector coverage than badminton/tennis/TT.",
 "WASB checkpoint C3 provenance remains an unresolved commercial-cleanliness blocker that gates BOTH options (head features and pseudo-labels) ? own-train weights or swap to a clean detector before commercial ship; a 'clean' model trained on provenance-unclear features does not launder the provenance.",
 "The forbidden-paid-LLM-teacher constraint materially raises the VLM path's cost: the entire public grounding-CoT data ecosystem (TVG-R1, etc.) is built on Gemini/GPT-generated CoT, which this project bans ? so the VLM path requires originating that supervision, a non-trivial cost that further justifies deferring it.",
 "'Very high precision/recall' should be defined operationally before the ship gate: current best cross-video LOVO ~0.615 (n=3, linear, threshold-transfer partly unsolved) is far from 'very high.' The path to high P/R is more data (n?dozens of matches across sports) + the segment-level TAL head, not a base-model swap. Set the target IoU threshold explicitly (e.g. F1@tIoU=0.5 for highlights, tighter for precise rally cuts).",
 "Qwen3-VL's grounding improvements over Qwen2.5-VL are claimed (explicit text-timestamp alignment) but not yet quantified on Charades-STA in public sources as of mid-2026 ? re-verify before any VLM-base commitment.",
 "Sequencing-discipline caveat per CLAUDE.md: this is a strategy recommendation; the committed next slice is the async job model (single-tenant). Building the event-index schema now is the no-regret action, but heavy VLM training stays gated on scaffolding + explicit owner go."
]
},
{
 "dimension": "Conversational Video-QA Architecture",
 "summary": "Extract once into a per-video event store; answer with a cheap query layer. Queries are structured-numeric (text-to-SQL + clip-cut), not semantic; most of the store exists. Build the bookkeeping now; punt the model until quality is solid.",
 "recommendation": "Adopt A: event store + session tables + a structured-query contract now; punt the NL parser/agent until quality beats ~0.583; reject B; Gemini only a fallback.",
 "confidence": "high",
 "options": [
 {


```

    "name": "A event store + query layer (RECOMMENDED)",
    "what": "extract once into play_segments+events; answer with SQL + VideoCompiler clip-
cut; LLM edge-only",
    "license": "clean: SQLite/ffmpeg/MIT LLM",
    "pros": [
        "process once query many; cheap; SQL exact where VLMs hallucinate; reuses existing
tables"
    ],
    "cons": [
        "extraction quality is the ceiling; open-ended needs a VLM fallback"
    ],
    "fit": "strong"
},
{
    "name": "B end-to-end VLM (rejected)",
    "what": "one model answers each turn; no store",
    "license": "paid/forbidden-as-trainer",
    "pros": [
        "simplest; open-ended native"
    ],
    "cons": [
        "expensive; bad at counting; license-hostile"
    ],
    "fit": "weak"
},
{
    "name": "C RAG over captions",
    "what": "captions to vector/graph DB; Graph-RAG",
    "license": "paid-model risk",
    "pros": [
        "best for open-ended Qs"
    ],
    "cons": [
        "cannot count exactly; future wing only"
    ],
    "fit": "medium"
}
],
"sources": [
    "NVIDIA AVAS",
    "VideoAgent",
    "AVA 2505.00254",
    "Google ADK",
    "database.py",
    "CODE_MAP.md"
],
"caveats": [
    "quality gates the feature (~0.583 LOVO)",
    "multisport semantics unvalidated",
    "single-tenant, owner-gated; fix the query contract early"
]
},
{
    "dimension": "Implementation language for the upcoming work (ML training+eval / platform /
eventual on-device), for a solo Python+FastAPI founder",
    "summary": "Recommendation: stay Python everywhere for the foreseeable roadmap, and treat
\"another language\" as a deferred, layer-local deployment detail ? never a rewrite. The three
layers each independently point to Python or to a language choice that is structurally decoupled
from your dev language, so there is no layer where a second language earns its keep against
solo-founder velocity right now.\n\n(1) ML TRAINING+EVAL is effectively Python-locked and that
is fine: the PyTorch/Transformers/ms-swift/ActionFormer/ASFormer stack is Python, and ms-swift
is invoked as a CLI (`swift sft/eval`) ? so even the orchestrator just shells out to a
subprocess. There is no non-Python alternative that retains native-video VLM training (the doc's
decisive finding) or your dependency-free numpy LogisticRegression Gen-0 seed. Locking here
costs nothing because training is batch/offline, not a latency hot path.\n\n(2) PLATFORM (async

```

jobs, multi-tenant API, storage/compute backends) has no real Python \"hot path\" to escape. The heavy bytes/GPU work already runs out-of-process (ffmpeg Windows-side, WASB in WSL, cloud GPU per-job via Modal/RunPod) ? Python is the conductor, not the orchestra. The control plane is I/O-bound (DB, signed URLs, job status, polling), exactly where the GIL never bound you and where Python 3.14 free-threading + asyncio now also remove the residual CPU-bound penalty. FastAPI 0.136+ officially supports free-threaded Python. fsspec/smart_open/Arq/Hatchet (all Python, all named in PLATFORM_ARCHITECTURE.md) cover storage+queue. A Go/Rust hot-path service would buy single-digit-ms wins on requests dwarfed by 40-minute GPU jobs ? pure overhead for a solo founder.

(3) ON-DEVICE is the one layer where another language eventually appears ? but it is DEPLOYMENT-only and structurally decoupled from your dev language by the export boundary. You train in Python, export to ONNX/Core ML/ExecuTorch/GGUF, and the on-device host (Swift for iOS Core ML, C++/Kotlin for ExecuTorch/Android, Rust via the `ort` crate) runs the artifact with zero Python dependency. ONNX Runtime alone exposes C++/C#/Java/Rust/JS APIs over the same model. So on-device does NOT dictate the dev language; it dictates a thin, separate, much-later runtime shell you write only if/when you ship a mobile compressed model. Per the owner's reframe, on-device is explicitly LATER (an upload-cost saver), so this is the correct layer to punt ? and punting it costs nothing architecturally because the export boundary is the seam.

Cross-cutting point on the conversational-QA north star and per-video bookkeeping: this REINFORCES Python-everywhere. The bookkeeping (rally inventory, shot counts, fault tags, timestamps, derived clips) is a durable relational store + retrieval/tool layer ? SQLite/Postgres + a thin query/agent layer ? which is bread-and-butter FastAPI/Python. Splitting language here would fracture the one codebase that has to hold per-video state, the eval harness, and the provider registry together.

When a second language IS justified (write it down, don't build it now): only at on-device ship time, and only the runtime shell (Swift/Kotlin/C++/Rust host around an exported artifact) ? never the trainer, never the control plane. Premature triggers to resist: rewriting the job queue in Go \"for throughput,\" or a Rust trajectory/feature-extraction service ? both are out-of-process or numpy-vectorized already; reach for them only if profiling on real load shows a Python CPU hot path that free-threading + a C-extension (numpy/Cython/PyO3-as-a-module) can't fix.

```
"options": [
  {
    "name": "Python everywhere (all 3 layers), on-device runtime shell deferred",
    "what": "Keep FastAPI control plane, Python ML training/eval (PyTorch + ms-swift CLI + ActionFormer/ASFormer), Python platform (fsspec/smart_open storage, Arq/Hatchet jobs, Postgres + repository layer). The eventual on-device model is a Python-trained artifact exported to ONNX/Core ML/ExecuTorch; the device-side host language is chosen ONLY at mobile-ship time and is a thin, separate shell.",
    "license": "All tooling MIT/Apache/BSD-compatible (FastAPI MIT, ms-swift Apache-2.0, ActionFormer/ASFormer MIT, fsspec/smart_open BSD/MIT, Arq MIT, ONNX Runtime MIT) ? clean under the never-corrupted guardrail.",
    "pros": [
      "Maximum solo-founder velocity: one language, one toolchain, one mental model across training, eval, platform, and per-video bookkeeping",
      "ML stack has no real non-Python option that keeps native-video VLM training (ms-swift) and your existing numpy LogReg Gen-0 seed ? staying Python avoids re-implementing the de-risking artifacts",
      "Conversational-QA bookkeeping (durable per-video state + retrieval/tool layer) is exactly FastAPI/Python's strength; keeps state, eval harness, and provider registry in one codebase",
      "Python 3.14 free-threading + asyncio + FastAPI 0.136 free-threaded wheels remove the historical GIL objection for CPU-bound endpoints if it ever bites",
      "Heavy work already runs out-of-process (ffmpeg, WSL WASB, cloud GPU) ? Python is the conductor, so 'hot path in Python' is largely a non-issue",
      "On-device export boundary (ONNX/Core ML/GGUF) means the device language is a future deployment detail, not a dev-language commitment now"
    ],
    "cons": [
      "On-device, when it ships, still requires writing SOME non-Python runtime shell (Swift/Kotlin/C++/Rust) ? Python doesn't run well on phones; this is unavoidable regardless of dev language",
      "Free-threaded Python is new (3.13 had ~40% single-thread penalty; 3.14 cut it to 5-10%) ? production maturity still settling, so treat as upside not dependency",
      "Python packaging/deploy ergonomics for a self-hosted serving stack are heavier than a single Go/Rust binary (mitigated by containers + Modal-per-job)"
    ]
  },
]
```

```

    "fit": "strong"
  },
  {
    "name": "Python core + Go/Rust hot-path service for the platform control plane / job
dispatch",
    "what": "Keep Python for ML + API surface, but extract job-dispatch / high-concurrency
endpoints / a trajectory-feature microservice into Go or Rust 'for throughput and a single
static binary.'",
    "license": "Go (BSD), Rust (MIT/Apache dual) ? both clean. ONNX `ort` Rust crate is
MIT/Apache. No license blocker; the objection is purely cost/benefit.",
    "pros": [
      "Single static binary deploy + lower memory footprint for the always-on control
plane",
      "Rust `ort` could serve an exported ONNX model with no Python in the serving path, if
a self-hosted inference tier ever needs maximal density",
      "Genuinely faster per-request latency on CPU-bound control-plane work"
    ],
    "cons": [
      "Premature for a solo founder: control plane is I/O-bound (DB, signed URLs, polling)
where Python was never bottlenecked; wins are single-digit-ms against 40-min GPU jobs",
      "Two languages = two toolchains, two CI paths, FFI/serialization seams, fractured per-
video-QA state logic ? directly taxes the velocity the owner needs",
      "No current Python hot path identified to justify it; heavy compute is already out-of-
process or numpy-vectorized",
      "If a Python CPU hot path ever appears, cheaper fixes exist first (free-threading,
numpy/Cython/Py03 C-extension) before standing up a second service"
    ],
    "fit": "weak"
  },
  {
    "name": "Rust/C++ on-device runtime shell at mobile-ship time (deferred, layer-local)",
    "what": "When the LATER compression goal ships a phone-side model, write only the
device-side inference host in the platform-native language: Swift+Core ML (iOS), Kotlin/C++
+ExecuTorch (Android), or Rust `ort`/C++ for a desktop power-user binary ? wrapping a Python-
trained, exported artifact.",
    "license": "ExecuTorch (BSD), Core ML (Apple SDK, app-distribution clean), ONNX Runtime
(MIT), llama.cpp (MIT). Clean ? but verify the BASE model's on-device license (note: Gemma 3
270M GGUF is commonly cited for on-device yet is license-REJECTED per your guardrail; stay on
Qwen3-VL-4B Apache-2.0 lineage).",
    "pros": [
      "Correct, unavoidable place for a second language: phones can't run the Python
trainer; the device needs a native host",
      "Fully decoupled by the export boundary ? does not touch training or control-plane
code, so it never fractures the main codebase",
      "Can be deferred with zero architectural cost today; the ONNX/Core ML export step is
the seam, planned for but not built now"
    ],
    "cons": [
      "Real engineering surface (mobile build, quantization validation, KV-cache mgmt) ? but
only when on-device is actually prioritized, which the owner explicitly defers",
      "Adds a maintenance target (app + model-version compatibility) ? budget it as a
distinct later phase, not now",
      "Ties up scarce solo-founder time the moment it starts; gate it behind proven quality
+ a measured upload-cost ROI"
    ],
    "fit": "medium"
  }
],
"recommendation": "Stay Python everywhere now, and explicitly write down \"another
language\" as a deferred, layer-local DEPLOYMENT decision ? not a rewrite and not a hot-path
split. Concretely: (a) ML training+eval stays Python (PyTorch + ms-swift CLI as a subprocess +
ActionFormer/ASFormer + your numpy LogReg Gen-0 seed); it is Python-locked and that costs
nothing because training is offline/batch. (b) Platform stays Python/FastAPI; do NOT introduce
Go/Rust for the control plane ? it is I/O-bound, the heavy work is already out-of-process
(ffmpeg/WSL/cloud GPU), and Python 3.14 free-threading + FastAPI 0.136 cover the only remaining

```

CPU objection. Use the Python-native pieces already chosen in PLATFORM_ARCHITECTURE.md (fsspec/smart_open, Arq?Hatchet, Postgres + repository layer). (c) On-device is the ONLY layer where a second language eventually appears, and it does NOT dictate your dev language: you train in Python and export to ONNX/Core ML/ExecuTorch/GGUF across the export boundary, then write a thin native runtime shell (Swift/Kotlin/C++/Rust-`ort`) ONLY when the LATER upload-cost-saver model actually ships. Until then, design the seam (an exportable model + a provider/ComputeBackend registry slot) and punt the shell. This keeps the conversational-QA per-video bookkeeping, the eval harness, and the provider registry in one Python codebase ? which is exactly what the quality-first, multisport, stateful-QA reframe needs. The single concrete language action now is to ensure the model architecture you train is export-clean (ONNX/Core ML-friendly ops) so on-device stays reachable without re-architecting ? a constraint on the model, not the dev language.",

```
"confidence": "high",
"sources": [
  "docs/PLATFORM_ARCHITECTURE.md (§3.2 ComputeBackend, §5 orchestration
  Arq/Hatchet/Temporal, §5A ms-swift training forge, fsspec/smart_open storage) ? existing Python-
  native posture",
  "docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md (Gen-0 ActionFormer/ASFormer MIT on local 2070,
  Gen-2 Qwen3-VL-4B Apache-2.0, Modal/RunPod cloud-per-job)",
  "docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md (ms-swift 'only native-video'
  finding; on-device VLM unbenchmarked; numpy LogReg Gen-0 seed)",
  "docs/CODE_MAP.md (heavy compute already out-of-process: ffmpeg Windows-side, WASB in
  WSL2; dependency-free numpy classifier.LogisticRegression; FastAPI app in backend/main.py)",
  "https://github.com/modelscope/ms-swift ? ms-swift is Python + CLI (swift sft/rlhf/eval),
  Apache-2.0, native video+audio training, 400+ MLLMs",
  "https://pypi.org/project/ms-swift/ ? ms-swift CLI/Python invocation surface",
  "https://onnxruntime.ai/docs/ ? ONNX Runtime cross-language APIs (C++/C#/Java/Rust via
  ort/JS): train in Python, run model with no Python dependency",
  "https://docs.rs/onnxruntime/ and ort crate ? Rust on-device serving of exported ONNX
  (YOLO-in-Rust example: Python-trained, Rust-served)",
  "https://www.edge-ai-vision.com/2026/01/on-device-llms-in-2026-what-changed-what-matters-
  whats-next/ ? ExecuTorch 1.0 GA (mobile), llama.cpp (desktop/proto), MLX (Apple); on-device
  runtime is a deployment-language choice, decoupled from training",
  "https://edgeaistack.ai/blog/edge-llm-runtime-stack-2026/ ? edge runtime stack
  (llama.cpp/ExecuTorch/MLX/vLLM) comparison; export-then-run model",
  "https://medium.com/@kevaldekivadiya2415/python-gil-vs-no-gil-real-fastapi-benchmarks-
  with-free-threaded-python-3-13-... ? FastAPI free-threaded benchmarks: CPU-bound ~8x up,
  I/O-bound unchanged (GIL never bound I/O)",
  "https://blog.imseankim.com/python-3-14-free-threading-jit-compiler-gil-removal-2026/ ?
  Python 3.14 cut free-threaded single-thread penalty to 5-10%, lock-free asyncio; FastAPI 0.136
  free-threaded support"
```

```
],
"caveats": [
  "The on-device second-language need is REAL but DEFERRED and layer-local: when a phone
  model ships you WILL write some Swift/Kotlin/C++/Rust runtime shell ? Python doesn't run on-
  device. The recommendation is to defer it via the export boundary, not to claim it never
  happens.",
  "Free-threaded Python (3.13t/3.14t) is the load-bearing reason the GIL objection is moot
  for CPU-bound endpoints; it is new and still maturing in production. Treat it as upside, not a
  hard dependency ? the platform is I/O-bound and fine on standard CPython regardless.",
  "One concrete language-adjacent constraint flows onto the MODEL (not the dev language):
  train with export-clean / ONNX/Core ML-friendly ops so the eventual compression target stays
  reachable. The tension that the immediate high-quality model isn't 8GB-constrained but must
  remain later-compressible is a model-architecture choice, handled in the model dimension, not
  here.",
  "An on-device base-model license note surfaced incidentally: Gemma 3 270M GGUF is
  frequently cited for on-device but is REJECTED by your guardrail (viral terms). Keep on-device
  bases on the Qwen3-VL-4B / Gemma-4 Apache-2.0 lineage ? a licensing item for the model
  dimension, flagged so the convenient on-device default isn't accidentally adopted.",
  "If profiling on real multi-tenant load ever shows a genuine Python CPU hot path in the
  control plane, escalate in this order BEFORE a second language: free-threading ->
  numpy/vectorize -> a C-extension via Cython/PyO3 (still a Python module) -> only then a separate
  Go/Rust service. No such hot path is identified today.",
  "ms-swift being Python-locked is not a risk: it is invoked as a CLI subprocess, so even a
  non-Python orchestrator would shell out identically ? the lock-in is to the PyTorch ecosystem,
```

which is industry-standard and unavoidable for VLM fine-tuning."

```
]
}
]
```

2. assistant (2026-06-09T11:56:29.765Z)

I'll review the recommendations against the actual repo state and guardrails. Let me verify the load-bearing claims against the source files.

3. user (2026-06-09T11:56:31.190Z)

```
1      # Owned-model implementation plan (DRAFT ? for owner review, no code started)
2
3      **Status:** PLAN ONLY. No owned-model implementation begins until the owner greenlights
a specific
4      milestone (per CLAUDE.md). Paid Gemini stays the always-ready primary/fallback
throughout.
5      **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` (the verified
study) +
6      `docs/archives/quality-iterations/2026-06-multivideo-lovo/` (the LOVO + prototype that
de-risk it).
7
8      ## North star & non-negotiables
9      - **Goal:** an OWNED, on-device rally?highlight model; local inference = the privacy
moat. The moat is the
10     consented dataset + eval harness, **not** the weights.
11     - **Guardrails (hard):** Apache-2.0/MIT bases only (Gemma-3 rejected); **never** train
on paid-LLM
12     (Gemini/OpenAI/Anthropic) outputs; exclude minors + separate training opt-in; split
data **by match**.
13     - **Sequencing discipline:** committed next platform slice is the **async job model
(scaffolding)**. Heavy
14     model *training* (Gen-2, cloud) waits for P0?P4 scaffolding + a healthy corpus +
explicit go. The
15     **corpus + data + compliance foundation is no-regret and can proceed first** ? it's
what makes later
16     training cheap and clean.
17
18     ## Why now is the right moment (today's evidence)
19     - Heuristic cross-video LOVO =  $0.443 \pm 0.279$  (Adarsh 0.722, Kushagra 0.163-ceiling).
The  $\pm 0.279$  is
20     *method* variance ? the unsupervised windowing is footage-fragile even when WASB
recall is excellent
21     (Kushagra: 97% of rallies have shuttle motion, yet unsegmentable).
22     - The learned prototype gets **held-out per-frame AUC 0.84?0.88** ; Kushagra oracle F1
**0.578 vs 0.163**.
23     ? The rally signal is *learnable*; the bottleneck is **data scale + calibration (n?5),
not method**.
24
25     ---
26
27     ## Phase 0 ? Foundation (start now; no GPU/scaffolding dependency)
28
29     ### M0.1 ? The labeled-corpus spine (the moat) . *owner-greenlight item*
30     - **What:** one canonical JSONL row per rally: `{video_id (file MD5), rally_ordinal,
sport, start, end,
31     ending_reason, label_type, source(human|pseudo|vendor), annotator/provenance,
consent/training_opt_in,
32     split(train|val|test, assigned BY MATCH), trajectory_ref, labeled_window}`.
33     - **Where:** authored in `sports-data-collector` (system-of-record; extends PR #13 + the
positioning doc);
34     the indexer only consumes transcoder outputs.
35     - **3 deterministic transcoders (one row ? three views, zero re-labeling):** (1)
heuristic/LogReg
```

```

36     candidate labels (already exists: `training.label_candidates`), (2) TAL ActivityNet-
JSON, (3) VLM clip?QA.
37     - **Guardrail enforcement:** provenance/consent/minors fields gate corpus membership
*before* transcoding;
38     Gemini labels never enter a training transcoder; the new `eval_partial_labels` window
flows into every view.
39     - **One scorer spine:** `metrics.match_intervals` for heuristic-labeling, TAL eval, and
the Gen-2 reward.
40     - **Acceptance:** round-trip test (collector export ? each transcoder ? indexer
loaders); split-by-match
41     enforced; no Gemini-sourced rows in any training view.
42
43     ### M0.2 ? Grow n?2 ? 5 distinct matches . *owner action + my scoring loop*
44     - **What:** tag ?3 more *distinct* matches (gBaaddy, TestLarge, + a small Roora batch).
I generate the
45     trajectory + score each as it lands (the loop we've been running).
46     - **"Healthy corpus" bar:** ?5 distinct match-sessions with full human labels ? stable
leave-one-match-out.
47     - **Acceptance:** n?5; per-match + LOVO numbers recorded; corpus diversity
(camera/court/lighting) noted.
48
49     ### M0.3 ? Compliance cleanup (do before it bakes in) . *owner-greenlight item*
50     - **(i) Purge Gemini-derived TRAINING labels.** The distilled pipeline currently trains
on Gemini silver
51     (`training.label_candidates` over Gemini CSVs) ? this **violates the no-paid-LLM-
training rule.** Retarget
52     training to human + open-teacher (WASB/own-model) labels only; Gemini stays
eval/teacher-reference.
53     - **(ii) WASB checkpoint provenance (C3).** The distributed
`wasb_badminton_best.pth.tar` is academic-data-
54     trained (provenance unresolved) ? it propagates to anything trained on its features.
**Decide:** document
55     as a hard Gen-0 prerequisite and plan ROADMAP Phase-4 "train own WASB weights," or
scope an alternative.
56     - **Acceptance:** no paid-LLM outputs in any training path; a written C3 decision +
prerequisite flag.
57
58     ### M0.4 ? Gen-0 A/B harness (built here; trains on your GPU box) . *owner-greenlight
item*
59     - **What:** productionize the prototype into a real harness ? feature stack (visibility-
run-length, velocity,
60     acceleration, net-crossing flags, smoothing/windowing) ? **ActionFormer (MIT, 1:1
rally=instance)** and/or
61     **ASFormer (MIT, TAS)** train?eval?**ship-gate**. Reuse `metrics.match_intervals`.
62     - **Ship gate (the honest bar):** held-out per-match F1 must beat the **honest cross-
video heuristic LOVO
63     (~0.44)** , not the fit-on-self 0.73. nanochat-style pass/fail verdict.
64     - **Where it runs:** training/eval on the **GPU/WSL machine** (can't run in this dev
container). Harness +
65     feature stack are authored here; you run the A/B.
66     - **Acceptance:** one-command train?eval?verdict; reproducible; the prototype's AUC
translated to segment F1
67     at n?5; documented A/B (ActionFormer vs ASFormer vs heuristic).
68
69     ---
70
71     ## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
72     - Run the A/B; if a learned Gen-0 clears the honest LOVO bar with per-video calibration,
**it becomes the
73     rally segmenter** (still fully local, MIT, on the 2070 Super). If not, the finding
(more data / better
74     features needed) is itself the next-step signal ? no over-fitting to declare victory.
75     - **Decision gate:** Gen-0 adopted ? more data/features first. Either way, the corpus +
harness persist.
76

```

```

77     ## Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + explicit go)
78     - **Gen-1 (Option A):** concatenate fully-local cue channels (audio-onset, pose/court)
onto the same head;
79     fusion in our rally layer. License/cost-vet the pose extractor (open item).
80     - **Gen-2 (Option B):** e2e multimodal VLM on an Apache-2.0 base (**Qwen3-VL-4B**
primary / Qwen2.5-VL-7B
81     fallback / **Gemma 4** end-state once a timestamp recipe is proven), SFT cold-start ?
GRPO/RFT with a
82     temporal-IoU-vs-golden reward. **Cloud training on Modal** (best no-access contract);
base license
83     re-verified per-checkpoint.
84     - **Hard pre-serving gate:** the **$/call crossover benchmark** (self-hosted-VLM vs
Gemini on real workload)
85     + an **on-device latency/throughput** measurement of the quantized base on the 2070
Super ? both currently
86     *unmeasured* and load-bearing. Don't stand up an owned serving stack until these clear
(COMMERCIALIZATION C7).
87
88     ---
89
90     ## Decisions I need from you (to unblock specific milestones)
91     1. **Greenlight which Phase-0 milestone(s) to start in code** (M0.1 corpus spine = my
recommendation; M0.3
92     cleanup; M0.4 harness). M0.2 (tagging) runs regardless.
93     2. **WASB C3 stance** ? accept "document as prerequisite" for now, or prioritize own-
WASB-weights sooner?
94     3. **Ratify "Apache-2.0/MIT"** (the study's approved list includes MIT bases/teachers;
the guardrail text
95     says "Apache-2.0 only").
96     4. **Corpus home** ? confirm the collector is the system-of-record for the spine
(consistent with PR #13).
97
98     ## Open prerequisites (carried from the study's critic ? none block Phase 0)
99     WASB C3 provenance · Gemma-4 license stays AMBER until a raw HF LICENSE check · on-
device VLM latency
100    unbenchmarked · real VLM VRAM/token cost on our 1080p/4K-60fps clips · DPA + no-train
for any cloud training ·
101    the owned-vs-Gemini $/call crossover volume.
102
103    ---
104    *This plan is design-only and reversible. Nothing here starts until you greenlight a
milestone.*
105

```

4. user (2026-06-09T11:56:31.287Z)

```

1     # CLAUDE.md ? project memory / agent handoff
2
3     Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
4     FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
5     repos: `sports-data-collector`, a VLC-based `rally-annotator`.
6
7     ## Read these first
8     - **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
9     contracts (the cold-start map; read before touching code).
10    - **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`,
`docs/PMF_MARKET_RESEARCH.md`,
11    `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also
`docs/ROADMAP.md`
12    (offline-segmenter narrative) and `docs/BACKLOG.md`.
13    - **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
14    **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). Archive policy: only move
15    **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level.
16
17    ## Current state (June 2026)

```

```

18 - **Scaffolding complete**; the **post-scaffolding strategy foundation** is laid and
merged
19 (#52/#53/#55/#57/#58). **No platform/owned-model implementation has started.**
20 - **Next slice = async job model (`DEFERRED_HARDENING_PLAN.md` #16), single-tenant.**
21 **Do not start implementation without explicit owner approval.**
22
23 ## Architecture in one breath
24 - **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers /
validators
25 / sports / annotations ? see `backend/pipeline/segmenters/base.py`. *Future:*
26 `StorageBackend` + `ComputeBackend` registries (PLATFORM §3).
27 - `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one
helper).
28 - Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
29 user_version` migration ladder in `backend/infrastructure/database.py`.
30
31 ## Committed guardrails (non-negotiable)
32 - **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
33 training data source** (terms/copyright). Training teachers/bases = **open Apache-2.0
only**.
34 - **Owned-model base:** Apache-2.0, video-capable (Qwen-VL / Gemma 4); **reject Gemma
3**
35 (viral license). **Exclude minors** from the training pool; per-user training data
needs a
36 separate explicit opt-in.
37 - **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly
future**
38 (GDPR Art. 9).
39 - **Generic data schema**; `coach ? athlete` is an **optional overlay**, not baked in.
40
41 ## Workflow conventions
42 - **Branch off `master`; ONE PR per work item; NO PR stacking. ** Don't open a PR unless
asked.
43 - Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.8+; `ffmpeg`
required.
44 - Never push to the default branch directly; descriptive commit messages.
45 - ?? This dev container lacks GPU / torch / cv2 / numpy ? the **full pipeline can't run
here**.
46 Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to
the owner.
47

```

5. assistant (2026-06-09T11:56:33.768Z)

Let me verify the remaining load-bearing claims against the actual files referenced as
"verified."

6. user (2026-06-09T11:56:35.298Z)

```
48: def match_intervals(preds: List[Interval], gts: List[Interval],
```

7. user (2026-06-09T11:56:36.820Z)

```

22:         """Initializes tables for videos, play_segments, and events."""
39:         CREATE TABLE IF NOT EXISTS play_segments (
61:             FOREIGN KEY (segment_id) REFERENCES play_segments(id) ON DELETE CASCADE
75:         CREATE TABLE IF NOT EXISTS candidate_segments (
91:             FOREIGN KEY (confirmed_segment_id) REFERENCES play_segments(id) ON
DELETE SET NULL
96:             ON candidate_segments(video_id, detector)
121:         "INSERT INTO play_segments (video_id, start_time, end_time, duration,
server, receiver, metadata) VALUES (?, ?, ?, ?, ?, ?, ?)",
153:         """INSERT INTO candidate_segments
171:         "UPDATE candidate_segments SET confirmed_segment_id = ? WHERE id = ?",

```



```

180:     def query_candidate_segments(self, video_id: str, detector: Optional[str] = None,
184:         query = "SELECT * FROM candidate_segments WHERE video_id = ?"
214:     def query_play_segments(self, video_id: str, filters: Dict[str, Any]) -> List[Dict[str,
Any]]:
222:         query = "SELECT r.* FROM play_segments r WHERE r.video_id = ?"

```

8. assistant (2026-06-09T11:56:43.255Z)

The repo claims check out: `match\_intervals` exists, `play\_segments`/`events`/`candidate\_segments` tables exist. Note the RECS claim "highlights becomes first-class per PLATFORM \$4.1" implies highlights is NOT yet a table ? let me confirm, plus check the LOVO number discrepancy and the current-state memory.

9. user (2026-06-09T11:56:44.309Z)

No matches found

10. user (2026-06-09T11:56:45.293Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-487 of 968 total (42181 tokens, cap 25000). Call Read with offset=488 limit=487 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```

1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-09 (PR review) · ? **OWNER REVIEWED PR #61 ? REFRAMED ROADMAP
(important checkpoint).** Owner left
13     ~12 inline comments (mostly state-updates + recommendations wanted). **Key roadmap
shifts to internalize:**
14     - **Immediate goal REFRAMED:** not "on-device model" but a **TRAINING FRAMEWORK +
HARNESS to produce a VERY HIGH
15     precision/recall model that extracts HIGHLIGHTS + RALLIES from a clip.** Quality
first. On-device = LATER (compression
16     for power users to save upload cost), definitely on roadmap but not now.
17     - **MULTISPORT FROM THE START** ? all net-separated racquet sports ? eventually all
sports; a SINGLE sports-agnostic model.
18     Collector + indexer already sport-generic by design. ? **ACTION: rename the repo off
"badminton" ASAP** (owner emphatic).
19     - **CONVERSATIONAL VIDEO-QA = north-star feature:** upload video ? ask contextual
questions ("longest rally", "make a clip
20     of all service faults", "rally >20 shots"). Needs per-video STATE/bookkeeping ? design
the framework for it. Owner asks:
21     embark now or punt + revisit after quality? (My lean: design the bookkeeping/event-
store layer now, punt the heavy
22     conversational model ? pending the strategy workflow.)
23     - **LICENSING GUARDRAIL (owner: "never corrupted")** EVERY dependency ? code, data, AND
weights ? must be usable in
24     PRIVATE PROPRIETARY COMMERCIAL software. **MIT / Apache-2.0 / BSD only.** Ratify (was
"Apache-2.0 only"). Enforce religiously.
25     - Owner wants a **coherent recommendation: local Gemma 4 vs ms-swift starting point**
(multisport, single, conversational,
26     commercial). NB: ms-swift = training FRAMEWORK, Gemma 4 = BASE model ? orthogonal;
clarify in the reply.

```

27 - Owner: ****upgrade WASB C3 (own-weights/provenance) to a first-class action item now.****
M0.2 (grow n) = practically DONE

28 (n=6); owner will add 1 more varied multi-court-club video soon ? Phase-0 ready to start.

29 - ****LANGUAGE QUESTION (owner callout, no preference):**** is Python right for the upcoming work, or brainstorm alternatives?

30 - ? ****STRATEGY WORKFLOW** launched (`wf\_f9aa1d1c-c83`)** to verify facts (Gemma4/Qwen/ms-swift licenses+capabilities) + give

31 coherent recommendations on the 4 Qs (model+starting-point, conversational-QA, multisport-impact, language). Then: rewrite

32 the plan/study coherently + reply to each PR comment + update state. README "3-match"? "6-match" already fixed (uncommitted).

33 - ?? ****NEXT:**** workflow lands ? coherent recs + roadmap rewrite (plan/study) + GitHub comment replies + single PR #61 update.

34

35 ****Checkpoint:**** 2026-06-09 . ? ****n=6:** MahadevpuraSingles added while owner reviews PR #61.** Owner moved ALL golden videos

36 + CSVs to `Annotation Setup/Collect/Golden Labelled/` (manifest label paths updated; trajectories unaffected in `%TEMP%`).

37 New: ****MahadevpuraSingles**** (singles, 1080p, 29.97fps, 31 rallies; traj `%TEMP%\mahadevpura\_traj.csv`, script

38 `run\_wasb\_mahadevpura.sh`). ****Borderline multi-court (contrast 2.2x)**** ? heuristic best 0.667, learned held-out ****AUC**

39 0.902/F1 0.712** (clean generalization to an unseen match). ****n=6 LOVO: heuristic 0.480 ± 0.242 vs learned 0.583**** (learned

40 holds ~+0.10 since n=3). Contrast metric now predicts heuristic F1 MONOTONICALLY across 5 matches (Mahadevpura's 2.2x?0.67

41 lands on the curve). ****Decoupling headline: learned held-out AUC 0.83?0.92 regardless of contrast; heuristic swings 0.16?0.75**

42 with it.** Committed to PR #61 (n=6 doc update). ? GOTCHA: don't combine inner `&` with run_in_background (orphans the job);

43 calibrate_local buffers output to end + is slow (~17?20min). "Ingested" = added to EVAL corpus (manifest); collector

44 import-local (system-of-record) still = M0.1, plan-only pending greenlight. ?? owner reviewing PR #61 (4 plan decisions pending).

45

46 ****Checkpoint:**** 2026-06-08 (later+4, SESSION WRAP) . ? ****n=5 corpus + MULTI-COURT root cause + heuristic discovery + SINGLE PR #61 for offline review.****

47 Owner tagged gBaaddy (47, 1080p, multi-court) + TestLargeVideo (6, short) ? corpus ****n=5**** (`output/human\_lovo\_manifest.json`).

48 - ****HEURISTIC DISCOVERY** (user asked "discover more heuristics like audio"): tested 2 label-free trajectory pre-filters

49 for multi-court (measured on all 5): foreground spatial-density gate = ****net ?0.04**** (helps multi-court +0.03, hurts clean

50 ?0.12); track-continuity/jump filter = ****no-op****. ? simple single-cue trajectory heuristics CAN'T isolate the foreground

51 court. ****AUDIO-foreground idea PROTOTYPED** (no more videos needed) ? WEAK ?:** on the 2 multi-court videos + clean

52 control, audio onset/RMS energy AUC only ****0.53?0.65**** for in-rally vs dead-time; HF bands (8?22kHz @48kHz) raise

53 CONTRAST (Kushagra 1.1?1.7x, gBaaddy 1.3?1.9x, confirming distance-attenuation physics) but AUC stays ~0.6. Fails:

54 dead-time isn't quiet ? foreground court loud during dead-time too (footwork/talk/pickup) + gym reverb ? audio loudness

55 ? court occupancy, not rally (same wall as E1 ~2.2x). ****Audio alone ? multi-court gate.**** Remaining heuristic lever =

56 real court-region detection (player-cluster/court-line, ROADMAP #2); robust path = the learned visual model (AUC 0.85).

57 All tested + corrected in PR #61 doc + a PR comment.

58 - ****COURT-REGION DETECTION SMOKE-TESTED ? LOW HEADROOM ?:**** only 12?14% of dead-time detections on the multi-court videos

59 fall OUTSIDE the foreground rally region (86?88% INSIDE) ? a perfect court gate lifts contrast only ~1.6?2.2x (still

60 failing). The confusion is the foreground court's OWN between-point activity, NOT separable background courts. So court

```

61      detection (heuristic OR own model) won't help. **3 heuristics now tested + FAILED
(spatial gate, audio, court
62      separability) ? all because it's a TEMPORAL rally-vs-non-rally problem in the same
space; only the learned model solves
63      it (AUC 0.83). NO cheap heuristic shortcut ? the learned segmenter is the path.** (In
PR #61 doc + PR comment.)
64      - ? **SINGLE PR #61** (https://github.com/avidullu/badminton-highlight-indexer/pull/61)
carries EVERYTHING for offline
65      review: rally_seq_proto + tests, owned-model STUDY + PLAN (4 decisions pending),
multivideo-lovo learnings (n=5 + multi-court
66      + heuristic discovery), CODE_MAP. Title/desc updated. Plan doc NOW committed (was
held). Temp experiment scripts deleted.
67      (#60 + collector #13 merged earlier; #61 = only open PR.)
68      - **? DEFINING FINDING (owner's insight, data-confirmed): heuristic fails on MULTI-COURT
footage.** gBaaddy (1080p like
69      Adarsh) heuristic best F1 **0.277**; Kushagra 0.163 ? both have BACKGROUND GAMES. WASB
tracks every shuttle incl. other
70      courts ? high dead-time visibility (gBaaddy 41%, Kushagra 57% vs clean Adarsh 18% /
doubles 22%). ***"Contrast" metric**
71      (in-rally÷dead-time vis) predicts heuristic success: ?3.6× works, ?1.9× fails. Academy
= multi-court = TARGET ENV ? these
72      are the most representative videos. **Lever = foreground-court isolation** (spatial
gate to prominent court; needs real
73      court-region detection, naive centroid muddled by foreground-shuttle-resting-low).
NEAR-TERM, no new model.
74      - **Learned prototype LOVO (n=5):** per-frame AUC 0.83±0.92 on every match; **recovers
BOTH multi-court videos**
75      (Kushagra 0.561, gBaaddy 0.574 vs heuristic 0.16/0.28). Kushagra climbs
0.049±0.417±0.561 with n. Mean 0.568 ? dipped
76      from 0.615 (n=3) ONLY due to the 6-rally TestLarge noise fold; **excl. it = 0.639**.
Per-video gap on hard videos is the signal.
77      - **Heuristic LOVO n=5 = 0.463 ± 0.247** (per-fold: Adarsh 0.751, doubles 0.758,
Kushagra 0.157, gBaaddy 0.295, testlarge
78      0.353). **FELL from n=3's 0.544** as multi-court videos entered ? the heuristic can't
represent the target env, so a more
79      representative corpus LOWERS its honest score. vs **learned 0.568** ? gap INVERTED
(heuristic led at n=2 +0.066; learned
80      leads at n=5 **±0.105** and widening). (b8jpxbryo was just slow, ~17min; completed
fine.)
81      - **SESSION WRAP shipped ? PR #61** (branch feat/owned-model-study-and-prototype): n=5 +
multi-court finding appended to
82      `docs/archives/quality-iterations/2026-06-multivideo-lovo/`; **session-handoff.md
REWRITTEN** (quality-iteration phase);
83      **CODE_MAP updated** (eval_partial_labels + rally_seq_proto + multi-court gotcha).
Owned model still PLAN-ONLY
84      (`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` DRAFT, 4 decisions pending).
85      - ?? **NEXT (tomorrow): owner tags more varied videos** ? re-run LOVO+prototype.
**Foreground-court isolation experiment**
86      = #1 lever. Owner reviews plan + greenlights a Phase-0 milestone (rec M0.1 corpus
spine). Commit plan doc on approval.
87
88      **Checkpoint:** 2026-06-08 (later+3) . ? **n=3 CROSSOVER ? learned model OVERTAKES the
heuristic (the data-backed green
89      light).** User tagged a 3rd distinct match **LargeTestVideo (DOUBLES, 5.3K±1920,
29.97fps, 49 rallies)** ? corpus now
90      n=3 (2 singles + 1 doubles, 3 cameras/frame-rates). Trajectory
`~/clips/largetestvideo_traj.csv` (script
91      `run_wasb_largetestvideo.sh`). Manifest `output/human_lovo_manifest.json` now n=3.
92      - **Single-video LargeTest (doubles):** heuristic best F1 **0.750** (tuned, R0.918),
best-fit 0.827 ? **doubles is
93      heuristic-FRIENDLY; Kushagra remains the LONE heuristic failure** (not a play-type
issue).
94      - **Honest LOVO, heuristic vs learned, n=2?n=3:** heuristic mean held-out **0.443±0.544
± 0.278** (Adarsh 0.725,
95      **Kushagra 0.151 capped at any n**, doubles 0.755). Learned prototype mean

```

****0.377?0.615**** (Adarsh F1 0.701/AUC 0.878,
 96 ****Kushagra 0.049?0.417****/AUC 0.850, ****doubles 0.729/AUC 0.914**** trained only on the 2
 singles). ****? The learned model**
 97 OVERTAKES the heuristic at n=3 (0.615 vs 0.544); at n=2 it trailed (0.377 vs 0.443).******
 Crossover predicted by the plan.
 98 - ****The story (honest):**** learned path SCALES with data (+0.24 for one match,
 calibration gap closing ? Kushagra
 99 0.049?0.417) and GENERALIZES across play type (doubles held-out AUC 0.914); heuristic
 is HARD-CAPPED on footage it
 100 can't represent (Kushagra ~0.15 at any n). Honest "bar to beat" is now ~0.54 (n=3),
 not the fit-on-self 0.73. ?
 101 Reusing every labeled video for training validated; bottleneck = data scale +
 calibration (n?5), not method.
 102 - ****Docs updated ? pushed to indexer PR #61**** (branch feat/owned-model-study-and-
 prototype): n=3 findings + final quality
 103 metrics + conclusions in `docs/archives/quality-iterations/2026-06-multivideo-lovo/`;
 study + index updated. (PR #60
 104 MERGED earlier; #13 MERGED.) ? Prototype speed features NOT yet fps-normalized (30 vs
 60fps) yet doubles still
 105 generalized (AUC 0.914) ? known refinement.
 106 - ****OWNED-MODEL: PLAN-ONLY per owner**** ? `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`
 written (DRAFT, uncommitted, for
 107 review). User chose "just plan, don't code yet." 4 decisions pending (which Phase-0
 milestone to greenlight [rec M0.1
 108 corpus spine], WASB C3 stance, ratify Apache-2.0/MIT, confirm collector = corpus
 system-of-record). NO owned-model
 109 code until greenlight. Momentum = tag more matches (n?5) + I score each.
 110 - ?? ****NEXT:**** owner reviews plan + greenlights a Phase-0 milestone; tag more matches ?
 re-run LOVO+prototype (expect
 111 learned mean to keep rising). gBaaddy/TestLargeVideo still untagged test videos
 (trajectories ready). Commit the plan
 112 doc on approval.
 113
 114 ****Checkpoint:**** 2026-06-08 (later+2) . ? ****Multi-video eval corpus + owned-model**
 research underway.****** User fully tagged
 115 Adarsh-and-Avi (40? ****96 rallies****) + provided ****Kushagra**** (32 rallies, 4K) for the
 TRAINING mix, and ****gBaaddy****
 116 (new, 1080p, untagged) + TestLargeVideo as held-out TEST videos. Generated full-video
 WASB trajectories (single 8GB
 117 GPU, serialized via wait-loops): `~/clips/{adarsh\_avi\_full,kushagra,gbaaddy}\_traj.csv`
 (+ %TEMP% copies). Scripts:
 118 `~/clips/run\_wasb\_{avi\_full,kushagra,gbaaddy}.sh`.
 119 - ****Adarsh-full eval (96 rallies, IoU?0.5, window [0,1405.8s]):**** baseline F1 0.632,
****tuned+gate2 0.727**** (P0.80/R0.67,
 120 best out-of-box), sweep ****ceiling 0.750**** (vel0.02/merge1.0/min_dur2.0/gate off, fit-
 on-self). vs the opening-10-min
 121 (40 rallies): best 0.667 / ceiling 0.742. ****Best-config REGION is identical across the**
 slice and full match******
 122 (merge_gap 1.0, min_dur 2.0; velocity irrelevant; gate-off=recall, gate-on=precision)
 ? early sign tuning may
 123 transfer; LOVO is the real test. Per-rally: `output/adarsh\_avi\_full\_vs\_human.csv`.
 124 - ? ****KEY FINDING ? Kushagra: WASB tracks shuttle BETTER than Adarsh (97% of GT rallies**
 have ?2 net-crossings, med 5,
 125 310 pts/rally vs Adarsh 76%) BUT no velocity-windowing config can segment it (per-
 video best-fit F1 only 0.163; baseline
 126 0.000; mean boundary err ~5s).****** Signal is PRESENT; the unsupervised heuristic FAILS ?
 likely because Kushagra's shuttle
 127 is visible 67.5% (vs Adarsh 40%), i.e. tracked through inter-rally activity ?
 boundaries bleed. Root cause is the
 128 heuristic class, NOT WASB recall or labels or tuning.
 129 - ****First cross-video LOVO (`calibrate\_local`, Adarsh-full 96 + Kushagra 32, human**
 labels, manifest
 130 `output/human\_lovo\_manifest.json`):****** baseline mean held-out F1 ****0.316**** ? calibrated
****0.443 ± 0.279**** (overfit gap
 131 +0.027, tiny). Per-fold: held-out ****adarsh 0.722**** (cfg tuned on Kushagra ? Adarsh's

own 0.777 ? config TRANSFERS great),

132 held-out ****kushagra 0.163**** (cfg tuned on Adarsh; = Kushagra's own ceiling).
****Asymmetry is the story:**** tuning transfers

133 fine WHERE the heuristic can solve the video; Kushagra is fundamentally unsegmentable
by velocity-windowing (ceiling

134 0.16) regardless of config. The ± 0.279 std = method variance, not config. ?
****definitive, data-backed case for the**

135 learned OWNED model****** (trajectory carries the signal; heuristic can't exploit it).
Kushagra labels+traj = valuable

136 TRAINING data even though the current pipeline can't use them.

137 - ? LOVO honesty: 1 entry per DISTINCT MATCH (group Adarsh-main + Adarsh-last-game as
one unit; calibrate_local has no

138 grouping ? keep matches as separate entries only). gBaaddy traj still inferring (test
video).

139 - ? ****PROTOTYPE** built (user asked) ? `backend/eval/rally\_seq\_proto.py` (+3 tests):******
learned per-frame TAS-style

140 segmenter over 6 hand-built WASB-trajectory features (vis-rate, speed, ****net-crossing**
rate******, spreads) ? pure-Python

141 LogisticRegression, honest LOVO. ****Held-out per-frame AUC 0.878 (Adarsh) / 0.841**
(Kushagra).****** Kushagra: LOVO-threshold

142 F1 0.049 (threshold doesn't transfer at n=2) but ****oracle-threshold F1 0.578 = 3.5x**
the heuristic's 0.163 ceiling.******

143 ? PROVES the rally signal is LEARNABLE where the heuristic fails; gap = threshold
CALIBRATION not representation; answers

144 the study's "can low-dim WASB features feed a TAS head?" = YES. Seed of Gen-0.

145 - ? ****OWNED-MODEL STUDY** done (11-agent workflow `wf\_49aab888-e10`, 35 sources) ?
`docs/archives/research/OWNED\_MODEL\_TRAINING\_STUDY.md`.******

146 3 generations off ONE corpus: Gen-0 ASFormer(MIT)/ActionFormer over WASB features ?
Gen-1 cue fusion ? Gen-2 e2e VLM

147 (Qwen3-VL-4B / Qwen2.5-VL-7B / Gemma-4, ALL Apache-2.0; Gemma-3 REJECTED).
****Platforms:** Modal primary (best written

148 no-access), RunPod-Secure fallback, Vertex on bench (India residency but tuning Gemini
? owned weights), 2070S =

149 Gen-0+inference only.****** Label-reuse = 3 transcoders, split-by-match. Critic caveats:
****WASB checkpoint provenance C3**

150 propagates****** (MIT code ? checkpoint provenance), Gemma-4 stay AMBER until raw license
check, ****purge Gemini-trained**

151 distilled labels****** to comply with eval-only policy. DESIGN-ONLY: no impl until P0-P4 +
healthy corpus + owner sign-off.

152 - ? ****DOC CHECKPOINT** shipped ? indexer PR #61****** (branch feat/owned-model-study-and-
prototype): prototype + study +

153 `docs/archives/quality-iterations/2026-06-multivideo-lovo/` learnings + index README.
Full suite ****254 green****.

154 - ? ****gBaaddy test trajectory inferring**** (started 13:13, ~16 min) ? ready for when user
tags gBaaddy.

155 - ?? ****NEXT:**** review/merge PR #61. User tags a held-out TEST match (gBaaddy/TestLarge)
? score calibrated config +

156 prototype on truly-unseen footage. ****Highest-ROI: grow n?2?5 distinct matches**** (small
Roora batch) ? honest LOVO +

157 per-video threshold calibration ? re-run prototype. Then (owner sign-off, post-
scaffold) corpus format + ASFormer Gen-0 A/B.

158

159 ****Checkpoint:**** 2026-06-08 (later+1) . ? ****3 work items shipped** (user said "go ahead
with all three").******

160 1. ****Tuning sweep**** ? added `--sweep` to `backend/eval/eval\_partial\_labels.py` (+test);
indexer branch `feat/eval-human-labels`

161 (commits 2c950f6 harness, 58d10cd sweep; NOT pushed/PR'd ? offer to PR). On Adarsh-
and-Avi: ****F1 0.667?0.742****

162 (velocity 0.03 / merge_gap 1.0 / ****min_dur 2.0**** / gate OFF; P0.67/R0.83). Dominant
lever = min_window_duration;

163 gate over-prunes this camera; velocity ~irrelevant. ? fit-on-self (one video,
optimistic) ? real number = LOVO.

164 2. ****Collector P0 normalizer + positioning doc ? PR #13****
(<https://github.com/avidullu/sports-data-collector/pull/13>,

165 branch `feat/manual-ingestion-normalizer`, pushed).

```

`src/cli/normalize_annotations.py` (+test) = standalone
166     pre-import guard: drops blank/?start/**negative-start** rows (else pydantic
validator/safe_float_required abort the
167     WHOLE import) + rennumbers fractional/dup rally_number 1..N by start (else
int(float())) PK-collides + DELETE-INSERT
168     destroys one). Also mm:ss?sec, pipe ending_reason?primary.
`docs/MANUAL_INGESTION_POSITIONING.md` = positioning +
169     schema map + P1/P2 roadmap. **Validated full chain on the REAL file vs a TEMP db**:
normalize 40?40 ? import-local
170     40 ? export 40-interval indexer CSV. Collector suite **124 green** (was 116).
171     3. **Adversarial review** (1 agent) caught a REAL hole pre-merge: negative start_time
was undefended (pydantic
172     `start_time must be non-negative` ? import abort) ? fixed (drop branch + test, commit
1fe1c24) + corrected 2 doc
173     inaccuracies (abort mechanism = pydantic validator not safe_float_required;
ending_reason is free-text not enum).
174     4. **Iteration doc archived + indexer PR'd:** `docs/archives/quality-
iterations/2026-06-human-gt-tuning/` (STATUS
175     COMPLETED; results table, sweep, failure modes, bugs, infra gotchas, reproduce) +
updated the quality-iterations
176     index README + 2026-06-initial seed forward-pointer. **Indexer PR #60 OPEN**
177     (https://github.com/avidullu/badminton-highlight-indexer/pull/60; branch feat/eval-
human-labels =
178     harness 2c950f6 + sweep 58d10cd + docs; 5 files, +446/-1, full suite 251 green).
179     - **GOTCHA learned:** import-local does NOT copy the video (records local_path); writes
obs report NEXT TO THE VIDEO
180     (clean those up after validation). DB path from `config/config.yaml` `database_path` ?
override via temp config for
181     side-effect-free validation (no `--config`/`--db` CLI flag exists).
182     - ? **BOTH PRs MERGED ? both repos CLEAN on master, no open PRs, branches deleted.**
Indexer **#60** (squash
183     `90fb8b3`: eval_partial_labels harness + sweep + iteration doc). Collector **#13**
(squash `099cf67`: normalize_annotations
184     guard + MANUAL_INGESTION_POSITIONING.md). NOTE: user said "review is merged" while
live API showed both OPEN ? I
185     completed both squash merges + synced.
186     - ?? **NEXT:** after user's 60-min tagging ? full-video re-run
(`backend/eval/eval_partial_labels.py`, window auto =
187     last labeled end) + (recommended) tag BREADTH across 3?4 videos for LOVO. P1 collector
follow-ups (labeled-window in
188     manifest, maturity ladder needs_review?confirmed?gold) when prioritized.
189
190     **Checkpoint:** 2026-06-08 (later) . ? **FIRST human-GT eval landed + collector manual-
ingestion positioning done.**
191     User did ~10 min of VLC rally-annotator tagging ? partial human CSV `Annotation
Setup/Collect/Adarsh and Avi.rallies.csv`
192     (40 rallies, covers 0?605s of a 23m29s video; schema
`rally_number,start_time,end_time,ending_reason,sport`).
193     - **Built `backend/eval/eval_partial_labels.py` (+test, 3 green) ? committed on branch
`feat/eval-human-labels` (commit
194     2c950f6, NOT pushed/PR'd).** Scores WASB rally segments vs a PARTIAL human CSV,
**restricted to the labeled window
195     [0, last_end]** so correct detections in the un-labeled tail aren't counted as FP (THE
key methodology for partial
196     labels). Reuses calibrate_wasb.make_predict_fn/load_golden_gts + rally_gate + metrics
+ export_wasb_segments.build_comparison.
197     - **Generated WASB trajectory for the labeled window** (first 610s ? 36,564 frames ?
`~/clips/adarsh_avi_traj.csv` +
198     Windows `%TEMP%\adarsh_avi_traj.csv`; WSL script `~/clips/run_wasb_avi.sh`; ~18 min
inference). ? GOTCHA: passing a
199     long multi-statement WSL command via the Bash tool with `run_in_background` MANGLES
`$VAR`/`${}` (empty expansion) ?
200     put the job in a .sh FILE (tr-strip CRLF into WSL) and background `bash file.sh`.
201     - **FIRST trustworthy human-GT numbers (IoU?0.5, 40 rallies in-window):** baseline F1
**0.650** (P/R 0.65/0.65);

```

202 tuned 0.623 (P0.50/R0.825, over-segments); **tuned+gate2 best F1 0.667**
 (P0.79/R0.575). Boundary err ~1s.
 203 Per-rally CSV `output/adarsh\_avi\_vs\_human.csv`. **Failure modes:** detector catches
 LONG rallies well (IoU 0.7?0.95,
 204 tight boundaries) but MISSES short 3?5s rallies (most of the 17 misses) and FRAGMENTS
 long ones (golden#1's 34.8s
 205 split into 2 sub-fires; most "FPs" are real-rally fragments + one possibly-untagged
 early rally @23.5s, NOT
 206 hallucinations ? precision is recoverable via merge_gap). gate2 alone hurt recall hard
 here (0.65?0.475) ? net-axis
 207 est may be off for this camera. Levers: min_window_duration? (short rallies),
 merge_gap? (fragmentation).
 208 - **Collector manual-ingestion positioning (ultracode 10-agent workflow
 `wf\_d251ad6c-472`, verdict confidence HIGH):**
 209 position sports-data-collector as the **single canonical system-of-record** ; indexer
 consumes ONLY intervals-only
 210 export (manifest.json + per-video start,end CSV). **Architecture ~80% there ? the lean
 .rallies.csv imports via
 211 `import-local` TODAY with ZERO code changes** (header matches DictReader,
 curate.py:232-298). **2 P0 landmines** ?
 212 thin pre-import normalizer: (a) blank/?start end_time RAISES and aborts the WHOLE
 import (parsing.py:22-28) ? drop bad
 213 rows; (b) fractional rally_number (7.5/16.5) truncates via int(float()) and PK-
 COLLIDES, DELETE-then-INSERT destroys
 214 one (db.py:217) ? renumber 1..N by start_time, keep orig in notes. DON'T widen PK to
 float (indexer ignores
 215 rally_number). End-state P1/P2: persist `labeled\_window\_start/end` in manifest (so
 indexer auto-restricts instead of
 216 inferring max-end ? WRONG when a video is tagged past its last rally); maturity ladder
 needs_review?confirmed?gold
 217 (? CurationStatus); provenance/annotation_source+timestamp; unify adapters + run Issue
 detection on manual imports;
 218 freeze intervals-only export contract. Rich `rally\_review.csv` (mm:ss, `rally` col,
 pipe reasons) NOT directly
 219 ingestible ? offline converter (P2, only if it's source of truth ? OPEN question).
 Enrichment rides in MANIFEST, never the eval CSV.
 220 - ?? **NEXT:** after the 60-min tagging ? re-run full-video (extract full frames + WASB
 + eval, window auto = max end of
 221 full labels). For LOVO (learned>baseline, n=2 blocker), recommend tagging BREADTH (3?4
 videos partially) over
 222 finishing one. Offer: P0 collector normalizer as a small PR;
 `docs/MANUAL\_INGESTION\_POSITIONING.md`.
 223
 224 **Checkpoint:** 2026-06-08 . ? **rally-annotator v1.3 ? PR #1 MERGED to `main`** (merge
 `d4c682a`, commit `b058262`,
 225 5 files +711/?127; local `main` fast-forwarded). Goal: in-plugin Help text +
 226 a documented ending-reason taxonomy + UX fixes the user asked for. Files changed in
 227 `C:\Users\avidu\Projects\rally-annotator`: `vlc/rally\_annotator.lua` (full v1.2?v1.3
 rewrite), `README.md`,
 228 `docs/CSV\_FORMAT.md`, `CHANGELOG.md`, + NEW `docs/ENDING\_REASONS.md`. **Installed to
 229 `%APPDATA%\vlc\lua\extensions\rally\_annotator.lua` for the user's live smoke test**
 (Reload extensions / restart VLC).
 230 - **VLC "About" finding (verified, high-confidence):** the empty "About ?lua" box the
 user screenshotted is VLC's
 231 Add-ons `AddonInfoDialog`; its "Lua script" body is a hardcoded constant from the
 addon scanner
 232 (`modules/misc/addons/fsstorage.c`) ? an extension CANNOT set it. descriptor
 `shortdesc`/`description` DO show, but
 233 only in the *Active Extensions* tab "More information" dialog. So real, author-
 controlled help = IN-DIALOG (Help
 234 button + `add\_html`). Enriched the descriptor too.
 235 - **v1.3 plugin changes:** ending reason now REQUIRED + non-sticky (placeholder `--
 choose reason --`; reset after
 236 save via dropdown `clear()`+rebuild ? VLC 3.0.x dropdowns have NO `set\_value`, first
 value after clear auto-selects);

```

237     two-step commit (Mark END *arms* ? Save Rally writes; button shows the # it will
write); editable Start/End text
238     fields; Recent-rallies `add_list` with Edit/Delete (uses `get_selection()`); Undo last
(cancel edit / clear mark /
239     pop last). Switched to an in-memory rows model + atomic CSV rewrite (tmp+rename,
Windows `.bak` rollback). CSV schema
240     UNCHANGED (`rally_number,start_time,end_time,ending_reason,sport`); forgiving parser
preserves extra columns.
241     - **Taxonomy (verified via a 3-agent workflow + adversarial check):** OUT-of-bounds =
the HITTER's error, forced ONLY
242     if pressured else unforced (NEVER auto-forced, never the opponent's winner); into-net
same; serve-into-net =
243     service_fault; rally net-cord-in = live (often a winner); serve net-cord diverges by
sport; default-to-unforced
244     tie-breaker. One fix the verifier caught and I applied: a badminton serve CAUGHT on
the net = service_fault (BWF Law
245     13.2), NOT a let. Full guide in `docs/ENDING_REASONS.md`; summary table in README.
246     - **Verification:** no Lua interpreter on PATH; adversarial static review (subagent) ?
state machine + VLC 3.0.x API
247     usage clean; fixed the 2 cross-platform bugs it found (POSIX `file://` path lost
leading `/`; `save_all` deleted the
248     live CSV before rename). ? STILL NEEDS the user's live VLC click-test (I cannot drive
VLC).
249     - **Follow-up (same session, also installed):** (a) RESUME ? on activate the plugin
reloads the video's existing
250     `.rallies.csv` (Recent list + continued numbering); `Refresh` now re-points to
whatever video is playing now and
251     loads its rallies (guarded against reload-during-edit so `edit_index` can't go stale;
playback *position* is NOT
252     restored). (b) UNDO clarity ? `Undo last` button shows the row it will remove (`Undo
last (#N)`) + a "Last row
253     (what Undo removes)" line in the status panel. README/CHANGELOG updated.
254     - **Reinstall gotcha (FIXED):** after merging PR #1 the user hit "plugin not reloading"
? the installed copy was
255     STALE (28251B, an earlier file-copy) vs merged source (28905B). Deleted + reinstalled
the merged file to
256     `%APPDATA%\vlc\lua\extensions\rally_annotator.lua`, hash-verified byte-for-byte
(28905B). **VLC 3.x "Reload
257     extensions" is unreliable for Lua code changes ? a full VLC restart is the reliable
path** (VLC was not running).
258     Lesson: always reinstall to the VLC dir AFTER the final edit, and verify by SHA-256,
not just byte count.
259     - ? **REAL root cause found (plugin never showed under View) ? fix PR #2 (v1.4) MERGED**
(merge `37a0ec1`,
260     commits `8d42a71` load-fix + `424eb3c` v1.4). The
261     stale file was a red herring; v1.3 NEVER loaded. **VLC scans an extension's
`descriptor()` in a RESTRICTED Lua
262     sandbox that does NOT expose base globals (`ipairs`, etc.).** v1.3 built
`REASON_ID`/`SPORT_ID` with **top-level
263     `ipairs` loops** ? scan-time error `attempt to call global 'ipairs' (a nil value)` ?
VLC silently skipped it. Fix:
264     move the table-building into `activate()` (full env); **NEVER call a global function
at the top level of a VLC Lua
265     extension.** (v1.2 worked only because its `ipairs` calls were all INSIDE functions =
activate-time.)
266     - ? **Reusable diagnosis method:** installed `lua5.1` in WSL ? `luac5.1 -p <file>`
(syntax) + load chunk in an EMPTY
267     env via `setfenv` (simulates the scan sandbox); and launched VLC with `--vv --file-
logging --logfile` + `--no-one-instance`
268     to read the actual `lua debug: Scanning Lua script ?` / `Error loading ?` lines. After
fix the log shows the scan with
269     `capability flags: 0x0` and NO error. WSL has no lua by default; `sudo apt install -y
lua5.1` adds `lua5.1`+`luac5.1`.
270     Also built a reusable **headless `lua5.1` harness** (`%TEMP%\ra_harness.lua`) that
STUBS the vlc dialog API + loads

```



```

271     the extension ? drives `activate()`/`show_help()`/`save_rally()` and asserts state ?
catches dialog-logic bugs
272     without VLC.
273     - ? **v1.4 (commit `424eb3c`, in PR #2):** (a) **Help toggle fixed** ? was "stuck"; VLC
didn't repaint the in-place
274     `set_text` on the shared status widget ? now a DEDICATED `add_html` panel removed via
`del_widget` (same trick).
275     (b) **Reason-stickiness fixed** ? selecting `winner` blocked the next rally also being
`winner`; VLC didn't repaint
276     the `clear()`-based dropdown reset ? now the reason RESETS to a savable **`unknown`**
default by RECREATING the
277     dropdown (del_widget+add_dropdown at the same cell; VLC has no `set_value`). Reason no
longer hard-required.
278     (c) **"Next rally #" field** ? resume/insert numbering anywhere; auto-advances to next
FREE number; Undo/Delete
279     re-sync it (no gaps). `unknown` added to README/ENDING_REASONS/CSV_FORMAT.
**Adversarial-review workflow (3 lenses)
280     caught 2 real gap bugs (undo/delete didn't re-sync Next#) ? fixed before commit.** ?
**VLC dialog lesson:**
281     in-place widget updates (`set_text`, dropdown `clear()`+re-add) DON'T reliably repaint
? use `del_widget`+re-add.
282     - ?? **ALL PRs #1/#2/#3/#5 MERGED ? `main` = `e10e543` (v1.5). Issue #4 CLOSED. SLATE
FULLY CLEAN:** only `main`
283     (local==`origin/main`), 0 other branches (all merged ones deleted+pruned
local+remote), 0 open PRs, 0 open issues,
284     working tree clean. Installed VLC file = v1.5.
285     - ? **v1.5 (merged via PR #5, closed issue #4):** (github.com/avidullu/rally-
annotator/pull/5, branch
286     `feat/playback-controls-and-tests`, commit `d8fc66f`). Issue #4 = user's "cognitive
enhancement" request: control
287     VLC playback from the plugin window (no window-switching). **Built: a `Back 5s / Play-
Resume / Pause / Fwd 5s`
288     row.** API **source-verified first** (2-agent workflow vs videolan/vlc-3.0):
extensions DO get `vlc.playlist`
289     (play/pause/status, registered in `extension.c`) + `vlc.var` seek on
`vlc.object.input()`. **Gotcha:** `pause()` is
290     a HARD TOGGLE (`playlist_TogglePause`) ? Pause/Play-Resume are **gated on
`vlc.playlist.status()`** so they never
291     flip wrong. Seek = relative  $\pm 5s$  via `vlc.var.set(input,"time", $\mu s$ )`, clamp 0, nil-
checked. Layout regrouped (playback
292     row at top; reason cell `(3,4)`?`(3,5)`, help row 14?15).
293     - ? **Committed test suite (answers the user's "test coverage?" question):** NEW
`test/dialog_test.lua` (run
294     `lua5.1 test/dialog_test.lua`, exit 0/1) ? stubs VLC dialog+playback API, loads the
real ext, **36 assertions**
295     (reason reset/consecutive/unknown, numbering+collision+undo-resync, help toggle,
**playback gating**, CSV bytes).
296     `test/README.md` + README Tests section. The throwaway `%TEMP%\ra_harness.lua` is now
superseded by this in-repo test.
297     Verified: 36/36 pass, `luac5.1 -p`, empty-env scan `SCAN_OK v1.5`, real VLC `-vv` log
(loads, `0x0`, no error).
298     Installed VLC file = v1.5. ? GUI render + REAL playback still need the user's live
click-through (untestable headless).
299     - ?? **NEXT:** nothing pending in the repo. User restarts VLC ? live-test playback +
layout  $\times 5$  sports; iterate on
300     feedback. Cheap future ideas floated: tweak seek step (5s?3s / add ?2s), per-sport
keyboard shortcuts (space=play/pause,
301     arrows=seek, S/E=mark). Run tests anytime: `lua5.1 test/dialog_test.lua` (36
assertions).
302
303     **Checkpoint:** 2026-06-07 (FINAL, session wrap) . ? **ALL MERGED ? observability
feature fully shipped; both repos
304     CLEAN.** Zero open PRs across all 3 repos (indexer/collector/obsreport); both repos on
`master`, 0 uncommitted, no
305     stray branches. Merged this session: indexer **`#41/#42/#43/#44`**, collector

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306     (private, v0.1.3).
307     - ##44 (last one)** = docs+tooling handoff: ``docs/CODE_MAP.md` REGENERATED** for the
#40 layered layout + config/
308         reporting subsystems (build-code-map workflow, 0 broken anchors, 325 lines) .
``run_all_tests.py`` at repo root
309         (one-command full suite; README points to it) . Grok coordination notes in
`docs/BACKLOG.md` + `LOW_REGRET_MOVES_PLAN.md`.
310     - ?? ``TWO config-migration bugs flagged for Grok`` (in BACKLOG + CODE_MAP $2.0 gotchas,
found by the regen, NOT fixed ?
311         Grok's config domain): (1) `/api/compile` ignores `--output-dir` (`OutputConfig` has
no `output_dir` field ?
312         always "output"); (2) `--output-dir` lives only on dynamic
`global_config._runtime_overrides`. Fix: add
313         `output_dir` to a config/runtime model + read consistently.
314     - ``Suites`` indexer 231, collector 116, obsreport 74 ? all green. Run via `python
run_all_tests.py`.
315     - ?? ``NOTHING PENDING for the observability work`` Optional future (not started): HTML
customer renderer; real
316         AI-provider timing (via Grok item #7); pin `obsreport` git+ssh tag once SSH/CI access
exists; extend report with
317         audio-readiness (Grok item #8). Original rally-detection modeling still PARKED pending
golden labels.
318
319
320
321     ``Checkpoint`` 2026-06-07 (latest+2) . ? ``FEATURE COMPLETE ? observability in both
tools, merged; Phase D done``
322     MERGED to master: obsreport (GitHub private, v0.1.3) . indexer ##42** (config fix) +
##41** (observability) .
323     collector ##10** (parser fix) + ##11** (observability). ``Debuggability eval done``
(15-agent Workflow: all 7
324     broken-run scenarios self-debuggable; fresh readers given only README/FAQ landed on the
root cause + cited the FAQ).
325     Refinements applied as follow-up PRs (OPEN, green, awaiting owner merge): indexer
##43** (README Q7 lists
326     wasb_hybrid/tracknet_hybrid; new fps Q14 + repoint fps_fallback off Q5; numeric fps
fallback; motion synthetic at
327     data level; silent_provider_noop segmenter?handoff + .env path; blur remediation; ``faq-
resolution golden test`` ?
328     231 green) and collector ##12** (frame-level evidence on delivery flags; review-only
framing for
329     possible_missed_rally/shot_density; ``faq-resolution golden test`` ? 116 green).
Eyeballed reports via rendered
330     HTML screenshot + customer summaries (clean, audience-appropriate). Eval artifacts:
`%TEMP%\obseval`.
331     ?? ``NEXT (when owner is back)`` merge #43 + #12 (both green, disjoint). Then the per-
video observability feature is
332     fully shipped across both tools + the shared obsreport lib. Optional future: HTML
customer report renderer (obsreport);
333     thread real AI-provider timing through the segmenters (currently 'not_captured'); pin
obsreport via git+ssh once SSH/CI access is set up.
334
335     ``Checkpoint`` 2026-06-07 (latest+1) . ? ``INDEXER OBSERVABILITY MERGED`` Both indexer
PRs merged to master:
336     ##42** (config fix `AppConfig.get` nested?dict; review nits addressed: dynamic section
scan + shim comment + tests)
337     and ##41** (per-video observability reports) ? master `98bb1d9`, ``229 tests green``,
feat branch deleted. Full
338     pipeline now sound end-to-end (validators run via #42; every run emits report via #41).
obsreport on GitHub (private,
339     tags v0.1.0?v0.1.3). ``Collector observability PR #11 STILL OPEN`` (rebased on #10, 113
green).
340     ?? ``NEXT: Phase D refinements (follow-up PRs) + golden tests`` Eval verdict: all 7
broken-run scenarios

```

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341     self-debuggable. Apply: (HIGH) repoint `fps_fallback_used` faq_ref off README#Q5
(keyframes) to a NEW fps FAQ; add
342     `wasb_hybrid`/`tracknet_hybrid` to README Q7 segmenter list. (MINOR) numeric fps
fallback `30 (fallback)`; motion
343     "synthetic" surfaced at data level; self-remediating blur line (name
`validation.min_blur_threshold`); collector
344     overlap frame-level detail; `possible_missed_rally` review-only framing. Add golden
tests: faq-ref-resolution (every
345     flag's cited anchor exists in its doc) + expected flag-set per fixture, in BOTH repos.
Indexer refinements ? new
346     branch off master; collector ? push to #11. Eval artifacts in `%TEMP%\obseval\`
(manifest.json + reports/ + gen_*.py).
347
348     **Checkpoint:** 2026-06-07 (latest) . ? **RE-SYNCEd onto parallel work + found/fixed a
config runtime bug.**
349     Indexer master advanced: **#39 (typed config backend/config/)** + **#40 (layered
directory restructure:
350     backend/pipeline/{validators,segmenters,stitcher,detectors,audio},
backend/infrastructure/database.py,
351     backend/api/static/)**. Rebased **observability PR #41 onto #40** (git rename-detection
merged my main.py reporting
352     onto the new imports; only repointed 2 test imports) ? **228 tests green**; #40 also
fixed the 2 config tests I'd
353     flagged. Collector observability **PR #11 rebased onto #10** (parser fix merged) ? **113
green**. obsreport **74 green**.
354     - ? **FOUND + FIXED a real config runtime bug ? indexer PR #42 (`fix/appconfig-nested-
get`)**. ** The typed AppConfig is
355     passed to validators/segmenters which use legacy chained
`config.get("validation",{ }).get("k")`; AppConfig.get
356     returned the nested *model* (no `.get`) ? **all 4 validators threw `ValidationConfig`
object has no attribute 'get`
357     on every real `python -m backend.main` run** (tests passed because they inject dicts).
Fix: AppConfig.get returns
358     nested sections via `model_dump` (deep dict-compat); +regression test; **215 green**;
CLI smoke now runs all
359     validators. **Surfaced BY the observability report** (nice dogfood). Owner approved
fixing it (its own small PR #42).
360     - **Open PRs now:** indexer **#41** (observability, green), indexer **#42** (config fix,
green), collector **#11**
361     (observability, green). obsreport on GitHub (private, tags v0.1.0?v0.1.3). Suggested
merge order: **#42 ? #41**
362     (so #41's runtime path is unbroken), and **#11** any time.
363     - ?? **STILL PENDING: Phase D** ? debuggability eval Workflow was running (fresh-reader
agents over 7 broken-run
364     reports); apply synthesized wording refinements + render a report to an image to
eyeball + add golden faq-resolution
365     tests, then push to #41/#11.
366
367     **Checkpoint:** 2026-06-07 (later) . ? **NEW FEATURE IN PROGRESS ? per-video
OBSERVABILITY REPORTS** across
368     BOTH tools (indexer + collector) + a NEW shared lib. Approved plan:
369     `C:\Users\avidu\claudelplans\drifting-snuggling-squirrel.md` (READ IT to resume).
Decisions: always-on,
370     reports written NEXT TO THE VIDEO by default (override via `report.output_dir`); BOTH
tools emit a technical
371     report (debuggable by a README+FAQ reader) AND a customer-friendly summary (strip
internals); shared schema via
372     a **new private repo `sports-obsreport`** (pkg `obsreport`) ? a reusable "report config
language."
373     **Research:** Soda/SodaCL = ELV2 (disqualified, not OSS);
GE/Evidently/ydata/Frictionless license-safe but
374     wrong shape ? built a small lib on permissive primitives (pydantic/Jinja2/PyYAML).
375     - ? **PHASE C DONE ? `sports-obsreport` lib built + LOCAL git tag `v0.1.0`** at
`C:\Users\avidu\Projects\sports-obsreport\`
376     (NOT on GitHub yet ? local only; editable-installed into Python 3.13 via `pip install

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-e . --no-build-isolation`).
377     71 tests green. Modules: `envelope.py` (typed ReportEnvelope = single source of
truth), `spec.py` (YAML config
378     language: sections + rules), `rules.py` (sub-Turing clause engine, no eval),
`render/{json_render,markdown}.py`
379     (technical + customer views from same envelope), `io.py` (next-to-video paths, atomic,
NEVER raises),
380     `recorder.py` (RunRecorder: in-flight stage capture, finalize?rules?verdict, write),
`diff.py` (stable_json for
381     cross-run diffs), `schema/report_envelope.schema.json` (contract + drift test).
specs/example.yaml ships
382     silent_provider_noop + fps_fallback_used. **Footgun caught:** correct build-backend is
`setuptools.build_meta`
383     (underscore) ? collector's pyproject has the hyphen typo `setuptools.build-meta` =
latent `pip install -e` break (flag/fix).
384     - ? **obsreport bumped to v0.1.1** (local tag) ? added optional `customer_message` on
flags/rules: customer report
385     shows ONLY curated flags (no technical leakage); technical always uses `message`.
Schema still 1.0 (additive).
386     - ? **PHASE A DONE (indexer)** on branch `feat/observability-reports` (NOT pushed;
commit 9dbc764). 222 tests green
387     (209 baseline + 13). Added `run_all_with_context` (backward-compat; run_all
unchanged); `backend/reporting/`
388     (soft-import wrapper ? no-op if obsreport absent + never-raises; pure `harvest.py`;
`spec.yaml` footgun rules);
389     wired `main.py` CLI + API in finally blocks; README FAQ Q10-Q13; requirements.txt
commented git+ssh dep. Verified
390     end-to-end via CLI smoke (report files appear next to video; customer summary clean;
technical findings cite FAQ).
391     - ? **PHASE B DONE (collector)** on branch `feat/observability-reports` (NOT pushed).
113 tests green. `src/reporting/`
392     (soft-import wrapper ? never raises; `adapter.py` Issue?Flag + delivery/import
recorders reusing rally_cvat;
393     `spec.yaml` sections + customer_verdict + item_noun=rally +
license_noncommercial/missing_fps rules). Wired
394     curate.py validate-delivery / convert-cvat / import-local; import report falls back to
annotations dir when video
395     absent (not cwd). docs/INGESTION_PIPELINE.md FAQ added. Fixed build-backend typo.
.gitignore ignores
396     *.report.json/.report.md/.summary.md. Verified via validate-delivery CLI smoke.
397     - ?? **obsreport bumped to v0.1.3** during B (local tags v0.1.0/.1/.2/.3): added
customer_message (curated customer
398     flags), customer_verdict (per-tool verdict phrasing), item_noun
(rally/point/highlight). 74 lib tests green.
399     - ? **DUP RISK RESOLVED (2026-06-07):** the collector `load_canonical_csv` falsy-zero
bug (drops a rally starting
400     at 0.0s) landed **exactly ONCE** via the dedicated branch `fix/canonical-csv-zero-
start` ? **PR #10 MERGED**
401     (merge commit `99dfa22`; fix `f7cab11` + regression test
`test_load_canonical_csv_keeps_rally_starting_at_zero`).
402     master shows the parser fix once ? **no double-merge**. The observability branch's
duplicate `fix(parser)` commit
403     was dropped, so collector **PR #11 (observability) carries NO parser fix** and merges
cleanly (doesn't touch
404     rally_cvat.py). Local fix branch deleted; remote auto-deleted on merge. (Originated
from spawned chip task_7a3d8081 +
405     this session's branch/PR ? same branch name, one PR.)
406     - ? **SYNCED TO REMOTE (2026-06-07):** obsreport ? github.com/avidullu/sports-obsreport
(PRIVATE, main + tags
407     v0.1.0?v0.1.3). Indexer ? **PR #41** (rebased onto #39 typed-config `backend/config/`;
main.py conflict resolved
408     = typed attr access for segmenter + my reporting; reporting normalizes
AppConfig?dict). Collector ? **PR #11**
409     (observability-ONLY, still OPEN; the dup parser-fix commit was DROPPED ? the dedicated
`fix/canonical-csv-zero-start`

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410     branch owned it as **PR #10, now MERGED to master** `99dfa22`).
411     - ?? **Indexer PR #41 has 2 PRE-EXISTING red tests from #39** (NOT mine, left untouched
to avoid clashing with Grok's
412     config work): test_config.py::test_load_default_config (config.json
default_segmenter=gemini vs model default
413     yolo_hybrid) + test_save_default_config_roundtrip (Path==str on Windows). My
reporting: 226 pass. Collector: 112 pass.
414     - ?? **NEXT: finish PHASE D debuggability eval** (ultracode Workflow: fresh-reader
agents over broken-run reports ?
415     refine wording; render a report to an image + eyeball) + committed golden tests
(faq_ref-resolution + flag-set) in both repos.
416     - **TODO (HELD for owner OK):** create private GitHub repo `sports-obsreport` + push
(tags v0.1.0/v0.1.1); then
417     pin obsreport via git+ssh in both repos' deps; then push indexer/collector branches +
open PRs.
418     - Phasing tasks tracked in the harness task list (C1-C4 + A done; B in progress; D
pending).
419
420     --- (prior checkpoint) ---
421     **Checkpoint:** 2026-06-07 . ? **SESSION SUNSET.** master clean (PRs #25?#35 merged; 209
tests green; only
422     feat/khelacharya-shot-intelligence retained; no open PRs). Audio E1 done ? classical
onset = ~2.2× ceiling, NOT
423     adopted (detector kept, PR #35); quantitative findings + lessons in
[[audio-e1-findings]] (+ plan §10). VLC
424     annotator = standalone repo github.com/avidullu/rally-annotator (MIT, multi-sport).
425     ***? RESUME NEXT SESSION on EITHER:** (1) **GOLDEN EVALS** ? owner self-rates clips in
the VLC annotator ?
426     `import-local`/`--labels` ? unblocks distilled-model LOVO (n?3) + human-gold E1 re-
confirm = the highest-leverage
427     unblock; OR (2) **AUDIO E2** ? learned timbre classifier (needs labeled hits). Also
queued: pose/stance #2, owner's
428     "other hunch", Option B (owned end-to-end model). Read [[audio-e1-findings]], [[code-
map-artifact]] (docs/CODE_MAP.md),
429     docs/RALLY_DETECTION_RESEARCH.md, ADR-007 to ramp.
430     ? **PR #36 (docs/AUDIO_E1_FINDINGS.md) MERGED to master 2026-06-07** (owner restored the
branch after an accidental
431     close; reopened + squash-merged). Record now in-repo + [[audio-e1-findings]] (memory) +
plan §10. master fully clean, no open PRs.
432
433     --- (prior checkpoint) CLEAN SLATE. Indexer PRs #25?#32 MERGED; master 202 tests green,
tree clean, branches pruned
434     (kept feat/khelacharya-shot-intelligence). Audio decided (ADR-007). VLC annotator ?
standalone public repo
435     github.com/avidullu/rally-annotator (MIT, multi-sport). **ALL PRs #25?#34 MERGED; slate
FULLY clean** (master only + retained feat/khelacharya-shot-intelligence; no open
436     PRs; tree clean). #33 removed the stale annotator copy; #34 expanded ADR-007 (fusion +
Option B) + repointed the
437     ref ? both merged safely. **PARKED: modeling waits on golden labels (user self-rates via
annotator); next code
438     lever = AUDIO (E1 go/no-go probe).**
439     **ARCHITECTURE DECISION captured (ADR-007 §Integration architecture, PR #34):** audio
fuses in OUR rally layer
440     (WASB stays frozen black box ? feature-level into distill_local + decision-level recall-
recovery); **Option B = the
441     intended end-state** ? once a healthy golden corpus exists, train an OWNED end-to-end
multimodal model (frames+audio+
442     pose+??rallies) fine-tuned on our data, dropping WASB-weight dependence + recall
ceiling. Modular cues now (audio?
443     stance?other) are how we earn Option B (each generates the labels it needs). See
resumption runbook ?.
444
445     ## DISTILLED MODEL BUILT (2026-06-06) ? user wants this approach; honest result = needs
more data
446     - **ALL PRs #27-#31 MERGED to master** (0c6f2c1). master now has: code-map,

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gate-A/rally_gate, gemini-refine
447     (+provider hardening + extract_video_chunk scale_width), calibrate_local,
distill_local. This session's 5
448     branches PRUNED (local+remote). Retained: feat/khelacharya-shot-intelligence (PR #11).
Pre-existing merged
449     stragglers still on origin (left, optional prune): docs/ending-reason-forced-
unforced(#19), feat/rally-slicer(#20),
450     feat/wasb-substrate(#21).
451     - **User will RATE the clips ? cached artifacts LEFT IN PLACE** (do NOT delete):
Temp\{adarsh,testlarge}_frames +
452     trajectories (~clips + Temp), Temp\verylarge_frames (partial 12246/46080).
Adarsh+TestLargeVideo trajectories
453     ready ? when user import-locals golden labels, calibrate_local/distill_local use them
with zero re-extraction.
454     (#30 stranded calibrate_local on gate branch ? RECOVERED in PR #31.)
455     - `backend/eval/distill_local.py` (PR #31, OPEN): learned model ? WASB recall-oriented
candidates (0.005,1.0,0.5)
456     ? features [duration, peak_velocity, mean_velocity, active_ratio, net_crossings]
(resolution/fps-INVARIANT)
457     ? classifier.LogisticRegression trained on Gemini labels (training.label_candidates
IoU). Honest LOVO + saves
458     output/local_rally_model.json. Reuses
classifier/training/rally_gate/metrics/gemini_refine.
459     - **HONEST RESULT (2 videos): LEARNED UNDERPERFORMS baseline** ? mean held-out F1 0.359
(learned) vs 0.438
460     (threshold baseline). Causes: (1) n=2 too few ? no transferable model; (2) WASB
proposal recall CEILING 0.43
461     on TestLargeVideo (its windows don't align with Gemini rallies; Adarsh ceiling 0.81).
Framework READY; needs
462     more teacher videos + better WASB recall to beat baseline. Final weights:
mean_velocity + duration dominate,
463     net_crossings barely used (unreliable at n=2).
464     - ? **VeryLargeTestVideo (4K, 46080 frames, 12m49s, 11.5GB) = held-out FINAL test for
USER MANUAL EVAL.**
465     Extraction running (task bzt8hc4ks, frames?1920). Then WASB ? baseline + learned rally
lists ? user manually
466     evals (manual = GT; no Gemini needed on it). User's manual eval becomes a 3rd-video
human GT ? strengthens model.
467     - PR #31 (feat/distill-local) OPEN ? merge to get distill_local + calibrate_local onto
master.
468
469     ## ? AUDIO SCAFFOLDING + PRELIMINARY E1 DONE (2026-06-07) ? PR #35 (feat/audio-hit-
detector-e1, OPEN). Verdict = AMBER.
470     - Built `backend/audio/hit_detector.py` (ffmpeg?numpy spectral-flux onset?adaptive peak;
NO scipy; 5 tests) +
471     `backend/eval/audio_e1_probe.py` (E1 diagnostic). 207 tests green. WASB untouched
(Option A).
472     - Ran preliminary E1 on EXISTING GoPro audio vs **Gemini SILVER** labels (no wait for
human labels): Adarsh (36
473     rallies) + TestLargeVideo (7).
474     - **RESULT (honest, AMBER):** ? audio fires in **100% of rallies** and **100% of WASB-
missed rallies have
475     hit-clusters** ? real RECALL potential (the bottleneck). ?? but **discrimination
weak**: 1.3x @k=2.5, peaks
476     **~2.8x @k=4**, then a CLIFF (k=6 ? ~0 hits); hit rate high (56/min @k=2.5 = many non-
shuttle transients:
477     footwork/voices/ambient); **audio-only F1 0.13-0.27 < WASB-only (0.33-0.54).** So
audio is NOT clean enough
478     to stand alone ? it's a **fusion/recall cue**, not a standalone segmenter.
479     - **OPTION A DONE (band-pass) ? NEGATIVE RESULT (2026-06-07, committed to PR #35):**
added a `band` knob to the
480     onset detector + swept low/mid/high bands x k at 32kHz on both videos. **Band-pass
does NOT help** ? HF
481     "shuttle-click" bands gave LOWER discrimination, not higher. **Ceiling stays ~2.2x**
(in-rally vs MID-MATCH

```

482 dead-time). ****Not a warm-up/label artifact**** (mid-match dead alone ~0.6 hits/s vs ~1.3
in-rally on both).
483 Root cause: between-point activity (shuttle tosses=our false-fire, footwork, talk) =
transients a classical
484 onset detector can't separate from rally hits. `k` only trades coverage for
discrimination. Kept the band knob
485 (cheap/optional) but it's NOT the fix. Full writeup: AUDIO_RALLY_DETECTION_PLAN.md
\$10.
486 - ****HONEST VERDICT: classical-onset audio is too weak to be a clean cue (~2.2x); band-
pass won't rescue it.****
487 Remaining audio path = ****E2 learned TIMBRE classifier**** (MFCC ? shuttle-thwack vs
toss/footstep/voice; needs

11. assistant (2026-06-09T11:56:54.085Z)

I have enough to assess. Two key facts confirmed: (1) `highlights` is NOT a current table (RECS correctly frames it as future per PLATFORM §4.1), and (2) the n=6 LOVO is ****0.583 learned / 0.480 heuristic****, while the implementation plan still cites older numbers (0.44 heuristic / 0.615 learned at n=3). The RECS use ~0.583 and ~0.54 inconsistently. Let me verify the PLATFORM_ARCHITECTURE ms-swift lean and the OWNED_MODEL_TRAINING_STUDY's "Apache-2.0 only / AMBER" framing to close the loop.

12. user (2026-06-09T11:56:54.289Z)

301:### Base model & training (Apache-2.0 only ? licensing is decisive)
336:as image sequences. ****Only the self-managed `ms-swift` explicitly advertises `native`**
343:| ****ms-swift**** (ModelScope) | Self-managed (rented/own GPU) | ****Native video+audio****;
Qwen3-VL/Omni, 300+ MLLMs | any incl ****7B**** | ? DPO/GRPO | ? full ownership | ****? video-QA;**
slots into BYO-compute (§3.2)****** |
356:- ****Early iteration / true video-QA ?**** self-managed ****`ms-swift`**** (or LLaMA-Factory /
380:reference for that **operability** ? one MIT repo, one `speedrun.sh` that runs
388:****Engine (don't reinvent):**** ****`ms-swift`**** already delivers this for VLMs ? CLI-driven,
397:2. ****Fine-tune**** via ms-swift on a ****ComputeBackend**** GPU (§3.2), run as an orchestration
404:So: ****golden labels ? ms-swift train (on a compute backend) ? eval gate ? register**
406:wrapper; ms-swift is the engine; the ****golden QA eval is the report card****.
561:[Omitted long matching line]

13. user (2026-06-09T11:56:55.078Z)

6:> Apache-2.0 / no-paid-LLM-training / privacy-local guardrails. Produced by an 11-agent
research
20:| ****Gen-0**** | now ? unblocks at n?5 matches | tiny ****temporal action segmentation (TAS)****
head over a hand-built per-frame feature stack from the WASB trajectory we already produce; per-
frame rally(1)/non-rally(0) ? merge into [start,end] | ****ASFormer (MIT, ~1.1M params)**** primary;
****ActionFormer (MIT)**** A/B challenger (overtakes once corpus = hundreds of matches) | ****local**
RTX 2070 Super**, trains in minutes, \$0 |
75:guardrail-clean): ****open-teacher pseudo-labels (WASB/own-model, confidence-gated ? never**
paid-LLM),
82:- ****Bases: Apache-2.0/MIT only.**** APPROVED: Qwen2.5-VL-7B/32B (Apache-2.0), Qwen3-VL-4B/8B
(Apache-2.0),
83: Gemma 4 E2B/E4B/12B-Unified (Apache-2.0 + native video/audio), InternVL3 ?38B (MIT/Apache),
ASFormer/
84: ActionFormer/TriDet (MIT). ****REJECTED:**** Gemma 3 (viral Gemma Terms + PUP), Qwen2.5-VL-3B
(NC research),
85: Qwen-72B/InternVL3-78B (100M-MAU cap), Llama Vision (700M-MAU), DAMO VideoLLaMA3 weights
(NC + paid-LLM-trained),
87: (Commons Clause ? use `sj-li/MS-TCN2` MIT). ? ****Gemma 4 size names =**
E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".******
88:- ****No paid-LLM training data**** ? Gemini/OpenAI/Anthropic = inference/eval/teacher-reference
only, ****never**** a
107:- ****?? WASB checkpoint provenance (C3) propagates.**** The study calls WASB "MIT," but that's
the **code**; the
116:- ****Policy ratifications needed:**** guardrail literally says "Apache-2.0 only" but the
approved list includes MIT ?

117: ratify "Apache-2.0/MIT". The MAU-cap rejections (72B/78B) may be over-conservative at our scale.

123:- ****Dropped option worth keeping:**** InternVideo2 encoder (MIT) + light grounding head ? the ***only*** data-realistic

14. assistant (2026-06-09T11:57:02.747Z)

I now have full confirmation of the key facts. One important catch: the STUDY (line 87) explicitly flags that ****Gemma 4 has NO plain "4B" ? sizes are E2B/E4B/26B-A4B/31B/12B-Unified****, and (lines 82-83) the STUDY's approved list says "Qwen3-VL-4B/8B" and "Gemma 4 ...native video/audio". The RECS claim "Gemma 4 caps video at ~60s with NO timestamp output" ? that contradicts the STUDY's own line 83 which lists Gemma 4 as "native video/audio" approved. Also the RECS reference "Gemma-4-4B" sizing loosely. Let me also note the STUDY ratifies the MAU-cap and approved-list nuances the RECS partly re-derive. I have enough to deliver the critique.

15. user (2026-06-09T11:58:08.120Z)

Output does not match required schema: root: must NOT have additional properties

16. assistant (2026-06-09T11:58:11.586Z)

I added an extra property. Let me resubmit with only the four required fields.

17. user (2026-06-09T11:58:55.156Z)

Structured output provided successfully

18. assistant (2026-06-09T11:59:04.445Z)

I've completed the completeness critique and returned it via StructuredOutput. Summary of what I verified and found:

****Verified against the repo (claims hold):**** ``metrics.match_intervals``, the ``play_segments`/`events`/`candidate_segments`` tables, and ``rally_seq_proto.py`` all exist as cited; ``highlights`` is correctly framed as a future (not-yet-existing) table; ms-swift is indeed the existing lean in PLATFORM_ARCHITECTURE.md.

****Most important findings:****

- ****One real guardrail inconsistency:**** the recs tell the owner to "clear the Gemma-4 AMBER" while elsewhere recommending counsel review, and while the project's own plan (line 99) and study keep it AMBER. Under "never corrupted," Gemma 4 should stay AMBER.
- ****A self-contradiction in evidence:**** the recs' Gemma-4 "60s/no-timestamp disqualifier" conflicts with the STUDY's own line 83 listing Gemma 4 as "native video/audio" approved (and line 87's "there is no plain 4B").
- ****Biggest gap:**** no cost/budget numbers anywhere for a solo founder, despite acknowledging 6-8 GPU GRPO and unverified VRAM.
- ****Stale/inconsistent metric:**** LOVO cited as both ~0.54 (n=3) and 0.583 (n=6); current verified state is n=6 learned 0.583 / heuristic 0.480, and the implementation-plan doc still carries the old n=3 numbers.
- The strongest part is the adversarial self-correction discipline (RacketVision retraction, forbidden Gemini-CoT datasets, WASB C3 not laundered, Gemini-silver purge).

The full structured breakdown (gaps, guardrail misalignments, weak claims, overall) was delivered through the tool.